

Ordinary Differential Equations: A Constructive Approach

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Chapter 1

Introduction

Why yet another book on ordinary differential equations? Information technology is having a transformative impact on engineering and science. Perhaps because of its historic ties with applications within mathematics this is especially evident in the area of analysis and in particular the subject of differential equations. Through the work of H. Poincaré on the three body problem [33, 32, 31, 8] already by the beginning of the previous century it was clear that solutions to nonlinear differential equations could contain fabulous complexity. However, it was through numerical simulations, e.g. the work of E. Lorenz [23], that chaos captured the imagination of the broader scientific and engineering communities. Tremendous theoretical advancements in the latter half of the twentieth century have provided us with deep insights into the structures of global solutions to nonlinear systems. Nevertheless, the typical practitioner begins and often *ends* their analysis of a differential equation with numerical simulations.

The fact that engineers, scientists, and even applied mathematicians are often satisfied with the results of simulations that do not provide guarantees of their validity can be seen as a testimony to the accomplishments of the numerical analysis community or as a point of serious concern for it is in the rare occasions of failure of well established and highly trusted methods that surprising discoveries can be found. Unfortunately, there have also been occasions where these failures have had serious consequences. Note that in the context of applications rigorous bounds on solutions is as important as knowledge of their existence. With this in mind the goal of this book is to provide a systematic set of techniques by which the practitioner can effectively obtain both rigorous bounds and proofs of existence of solutions. We hasten to add that this is a book on nonlinear analysis, it is not a book on numerical analysis. Our focus is on developing estimates and techniques that lead to proofs of the classical results in the theory of ordinary differential equations, but are amenable to input from numerical simulations.

For any text choices need to be made concerning the choice of material and its presentation, since the results and the philosophy presented in this book are relatively new we

focus on clarity of ideas as opposed to generality of results. A benefit of this approach is that the material is accessible with minimal background: undergraduate courses in analysis and linear algebra suffice. The cost is that standard elementary results in differential equations are introduced later in the book. Another cost is that most nontrivial results are presented in the setting of analytic nonlinearities as this allows us to make use of classical results from complex analysis.

However, many ordinary differential equation models are at least locally analytic and thus there is no shortage of examples and Second, clearly written proofs of the general results are available in other texts. Third, where available we provide references to the li

There are excellent modern references with different perspectives of the subject.

Chicone classical ordinary differential equations [10]

Robinson dynamical systems [34]

Perko applications [30].

We use Newton's method. Deuflhard [15]

Notation

Given $x, y \in \mathbb{R}^n$,

$$x \ll y \quad \text{if and only if} \quad x_k \leq y_k \text{ for all } k = 1, \dots, n.$$

The default norm on $\mathbb{R}^n$ is the 2-norm:

$$\|x\| \stackrel{\text{def}}{=} \sqrt{\sum_{k=1}^n |x_k|^2}.$$

Let $\|\cdot\|_X$ denote a norm on a space X . Then

$$\overline{B_\epsilon(x_0)} \stackrel{\text{def}}{=} \{x \in X \mid \|x - x_0\|_X \leq \epsilon\}$$

Let J be an interval and $\alpha: J \rightarrow X$ where X is a normed space with norm $\|\cdot\|_X$. The C^0 norm of α is given by

$$\|\alpha\|_{C^0(J)} \stackrel{\text{def}}{=} \sup \{\|\alpha(t)\|_X \mid t \in J\}.$$

Let $M_n(\mathbb{R})$ the set a $n \times n$ matrices with real coefficients and $M_n(\mathbb{C})$ the set a $n \times n$ matrices with complex coefficients.

Given a matrix $A = \{a_{i,j}\}_{i,j}$ and a vector $v = \{v_i\}_i$ (real or complex valued), then the elementwise absolute value is represented by $|A| = \{|a_{i,j}|\}_{i,j}$ and $|v| = \{|v_i|\}_i$, where $|\cdot|$ denotes the absolute value.

The *sup norm* on $\mathbb{R}^n$ is defined as follows. Given $x = (x_1, \dots, x_n) \in \mathbb{R}^n$

$$\|x\|_\infty \stackrel{\text{def}}{=} \max_{k=1, \dots, n} \{|x_k|\}.$$

Given a matrix A , the matrix norm induced by $\|\cdot\|_\infty$ is defined to be

$$\|A\|_\infty = \max_{\|x\|_\infty \leq 1} \|Ax\|_\infty.$$

If $T: X \rightarrow X$, then its n -th iterate T^n is recursively defined by $T^n(x) = T(T^{n-1}(x))$.

Part I

Finite Dimensions

Chapter 2

Existence of Zeros of Functions

Given an ordinary differential equation $x' = f(x)$ the simplest object of interest is an equilibrium, i.e., a point $\tilde{x}$ satisfying $f(\tilde{x}) = 0$. The focus of this chapter is on developing a particular technique that we call the method of radii polynomials for proving the existence of equilibria and obtaining bounds on their location. The philosophy of the strategy is quite simple. Transform the problem of finding a zero to the problem of finding a fixed point. Newton's method, discussed in Section 2.3, provides a conceptual approach for constructing a function T with the property that if $T(\tilde{x}) = \tilde{x}$, then $f(\tilde{x}) = 0$. Of course, this is only of use if there is a general method by which one can prove that T has a fixed point. For this we make use of a classical result, the Contraction Mapping Theorem that is presented in Section 2.1. This is an extremely powerful theorem in that it guarantees the existence of a unique equilibrium. As the name suggests the key requirement is that T be a contraction mapping.

Therefore, the challenge in implementing this strategy is identifying readily checkable conditions under which T is a contraction mapping. This is the role of the above mentioned radii polynomials. Based on the fact that given today's technology people study differential equations numerically we assume that $\bar{x}$, an approximate zero to f , has been proposed. As detailed in Theorems 2.4.2 and 2.5.2 the radii polynomial is derived from bounds associated with $\bar{x}$, f and T , and if this polynomial assumes negative values then an explicit domain on which T is a contraction mapping is provided. As a consequence one obtains the existence of a unique equilibrium and bounds on its location.

We hasten to add that there is an art to this strategy, and it involves the construction of the map T . As indicated above Newton's method provides the impetus and thus the reader may not be surprised to discover that throughout this text T takes the following form:

$$T(x) = x - Af(x).$$

The art is in the choice of a linear operator A . Most of the discussion in Sections 2.4 and 2.5 focuses on providing intuition on how to choose A so that analytic bounds that produce

effective radii polynomials can be achieved. We make use of this discussion to introduce a variety of models that will be used throughout this book as more sophisticated solutions and structures of differential equations are discussed.

We finish this Chapter with a theorem that indicates that if $\tilde{x}$ is a zero of f , $Df(\tilde{x})$ is invertible, the initial guess $\bar{x}$ is sufficiently close to $\tilde{x}$, and one has sufficient computational resources, then the radii polynomial approach guarantees that $\tilde{x}$ is an equilibrium.

2.1 Contraction Mapping Theorem

Consider a function $T: X \rightarrow X$ where X is a topological space. An element $\tilde{x} \in X$ is a *fixed point* of T , if $T(\tilde{x}) = \tilde{x}$.

In general finding fixed points of a function is a nontrivial task. An exception is when iterations of the function lead to a fixed point. Given $x_0 \in X$, inductively define $x_{n+1} := T(x_n)$ for $n \in \mathbb{N}$. From that definition $x_1 = T(x_0)$, $x_2 = T(x_1) = T(T(x_0)) =: T^2(x_0)$, and in general $x_{n+1} = T^n(x_0)$, where T^n denotes the composition of T n times with itself. A fixed point is *globally attracting* if $\lim_{n \rightarrow \infty} x_n = \tilde{x}$ for all $x_0 \in X$. The contraction mapping theorem, discussed below, provides a simple criterion that guarantees that a function has a globally attracting fixed point.

To effectively identify globally attracting fixed points it is essential that limits exist under reasonable bounds. This restricts the type of topological space of interest.

Definition 2.1.1. A metric space $(X, \mathbf{d})$ is said to be *complete* if every Cauchy sequence converges in X , i.e., if given any $\epsilon > 0$ there exists $N(\epsilon)$ such that $n, m > N(\epsilon)$ implies $\mathbf{d}(x_n, x_m) < \epsilon$, then there is some $y \in X$ with $\lim_{n \rightarrow \infty} x_n = y$.

Definition 2.1.2. Let $(X, \mathbf{d})$ denote a metric space. A function $T: X \rightarrow X$ is a *contraction* if there is a number $\kappa \in [0, 1)$, called a *contraction constant*, such that

$$\mathbf{d}(T(x), T(y)) \leq \kappa \mathbf{d}(x, y)$$

for all $x, y \in X$.

Theorem 2.1.3 (Contraction Mapping Theorem). *Let $(X, \mathbf{d})$ be a complete metric space. If $T: X \rightarrow X$ is a contraction with contraction constant κ , then there exists a unique fixed point $\tilde{x} \in X$ of T . Furthermore, $\tilde{x}$ is globally attracting, and for any $x \in X$,*

$$\mathbf{d}(T^n(x), \tilde{x}) \leq \frac{\kappa^n}{1 - \kappa} \mathbf{d}(T(x), x). \quad (2.1)$$

Proof. Choose $x_0 \in X$ and recursively define $x_{n+1} \stackrel{\text{def}}{=} T(x_n)$. By the assumption that T is a contraction,

$$\mathbf{d}(x_{n+1}, x_n) = \mathbf{d}(T(x_n), T(x_{n-1})) \leq \kappa \mathbf{d}(x_n, x_{n-1}).$$

Thus, by induction

$$\mathbf{d}(x_{n+1}, x_n) \leq \kappa^n \mathbf{d}(x_1, x_0).$$

Applying the triangle inequality, we have for $n < m$

$$\begin{aligned} \mathbf{d}(x_n, x_m) &\leq \sum_{j=n}^{m-1} \mathbf{d}(x_{j+1}, x_j) \\ &\leq \sum_{j=n}^{m-1} \kappa^j \mathbf{d}(x_1, x_0) \\ &\leq \kappa^n \left(\sum_{k=0}^{\infty} \kappa^k \right) \mathbf{d}(x_1, x_0) \\ &\leq \kappa^n \frac{1}{1 - \kappa} \mathbf{d}(x_1, x_0). \end{aligned} \tag{2.2}$$

This implies that $\{x_n\}$ is a Cauchy sequence. Since X is complete there exists $\tilde{x} \in X$ such that

$$\lim_{n \rightarrow \infty} x_n = \tilde{x}.$$

By continuity of T ,

$$\tilde{x} = \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} T(x_{n-1}) = T\left(\lim_{n \rightarrow \infty} x_{n-1}\right) = T(\tilde{x}).$$

This establishes the existence of a fixed point.

We prove by contradiction that the fixed point is unique. Assume $\tilde{x}$ is as above and $\tilde{y}$ is another fixed point of T , that is $T(\tilde{y}) = \tilde{y}$ and $\mathbf{d}(\tilde{y}, \tilde{x}) > 0$. Then

$$\mathbf{d}(\tilde{y}, \tilde{x}) = \mathbf{d}(T(\tilde{y}), T(\tilde{x})) \leq \kappa \mathbf{d}(\tilde{y}, \tilde{x}).$$

Dividing by $\mathbf{d}(\tilde{y}, \tilde{x})$ gives $\kappa \geq 1$, which is the desired contradiction.

Turning to (2.1), note that since $x_0 \in X$ was arbitrary, the estimates of (2.2) are independent of the base point x_0 . Applying the triangle inequality we have

$$\begin{aligned} \mathbf{d}(T^n(x), \tilde{x}) &\leq \mathbf{d}(T^n(x), T^m(x)) + \mathbf{d}(T^m(x), \tilde{x}) \\ &\leq \frac{\kappa^n}{1 - \kappa} \mathbf{d}(T(x), x) + \mathbf{d}(T^m(x), \tilde{x}). \end{aligned}$$

for any $x \in X$, and all $m > n$. Taking the limit as $m \rightarrow \infty$ yields the bound claimed in (2.1). Finally, (2.1) and the fact that $\kappa \in [0, 1)$ proves that $\tilde{x}$ is a globally attracting fixed point. $\square$

Remark 2.1.4. Observe that given a contraction mapping $T: X \rightarrow X$, the rate at which points in X converge to the globally attracting fixed point $\tilde{x}$ is determined by κ . In particular, the smaller κ is, the faster iterates under T converge to $\tilde{x}$.

2.2 Mean Value Theorem

In this section, we introduce the mean value theorem and its corollary, the mean value inequality, which, when combined with the contraction mapping theorem, provides a strategy to obtain computer-assisted proofs of existence of solutions of zero finding problems using a Newton-like operator.

Recall that a *Banach space* X is a complete normed vector space and that the norm $\|\cdot\|$ defines a metric $\mathbf{d}$ on X by

$$\mathbf{d}(x, y) := \|x - y\|.$$

For purposes of this book we always work in the setting of Banach spaces or subsets thereof. Since the contraction mapping theorem is applicable in the setting of arbitrary complete metric spaces it is clearly applicable in any closed subset of a Banach space. We leave it to the reader to check that $\mathbb{R}^n$ and $\mathbb{C}^n$ are Banach spaces.

Standard norms on $\mathbb{R}^n$ or $\mathbb{C}^n$ are the *1-norm*

$$\|x\|_1 := \sum_{k=1}^n |x_k|,$$

the *2-norm* or *Euclidean norm*

$$\|x\|_2 := \left(\sum_{k=1}^n |x_k|^2 \right)^{\frac{1}{2}},$$

and the *∞ -norm* or *sup norm*

$$\|x\|_\infty \stackrel{\text{def}}{=} \max_{k=1, \dots, n} \{|x_k|\}.$$

A fundamental result (see Proposition 19.2.4) is that these norms induce the same topology on $\mathbb{R}^n$ ($\mathbb{C}^n$) and thus we are free to choose the most convenient norm for our purposes. With this in mind when working with a finite dimensional vector space we denote the norm by $\|\cdot\|$ unless we make use of specific properties of a particular norm.

Definition 2.2.1. Let $(X, \|\cdot\|)$ be a normed linear space and let $A: X \rightarrow X$ be a linear map. The *operator norm* on A is given by

$$\|A\| := \sup_{x \in X \setminus \{0\}} \frac{\|Ax\|}{\|x\|} = \sup_{\|x\|=1} \|Ax\|.$$

The following proposition the proof of which is left to the reader provides two fundamental properties of the operator norm.

Proposition 2.2.2. *Let $(X, \|\cdot\|)$ be a normed linear space and let $A, B: X \rightarrow X$ be linear maps. Then,*

$$(i) \|Ax\| \leq \|A\|\|x\| \text{ for all } x \in X.$$

$$(ii) \|AB\| \leq \|A\|\|B\|$$

We also leave to the reader to prove the following result.

Proposition 2.2.3. *$M_n(\mathbb{R})$ and $M_n(\mathbb{C})$ are Banach spaces under the operator norm.*

Remark 2.2.4. It is important to note that the choice of norm $\|\cdot\|$ has an impact on the associated operator norm. Let $M_N(\mathbb{R})$ and $M_N(\mathbb{C})$ denote the set of $N \times N$ matrices over $\mathbb{R}$ and $\mathbb{C}$, respectively. Given a matrix $A = [a_{i,j}] \in M_N(\mathbb{C})$ (or equivalently $A \in M_N(\mathbb{R})$), the associated matrix norms are

$$\|A\|_1 = \max_{1 \leq j \leq N} \sum_{i=1}^N |a_{i,j}| \quad (2.3)$$

$$\|A\|_2 = \sqrt{r_\sigma(A^*A)} \quad (2.4)$$

$$\|A\|_\infty = \max_{1 \leq i \leq N} \sum_{j=1}^N |a_{i,j}| \quad (2.5)$$

where A^* is the conjugate transpose of A and $r_\sigma(A^*A)$ denotes the maximum of the magnitudes of the eigenvalues of A^*A . Observe that if the matrix A is known explicitly, then computing $\|A\|_1$ or $\|A\|_\infty$ is much easier than determining $\|A\|_2$.

We are ready to state and prove the main results of this section.

Theorem 2.2.5 (The mean value theorem). *Suppose that $U \subset \mathbb{R}^n$ is open and that $f: U \rightarrow \mathbb{R}^n$ is C^1 . Consider $x, y \in U$ such that the line segment*

$$(1-t)x + ty, \quad t \in [0, 1]$$

is contained in U . Then,

$$f(y) - f(x) = \left(\int_0^1 Df((1-t)x + ty) dt \right) (y - x),$$

where Df denotes the Jacobian of f .

Proof. Denote $f = (f_1, \dots, f_n)$ and fix $i \in \{1, \dots, n\}$. Define $g_i: [0, 1] \rightarrow \mathbb{R}$ by

$$g_i(t) = f_i((1-t)x + ty).$$

Hence,

$$\begin{aligned}
 f_i(y) - f_i(x) &= g_i(1) - g_i(0) \\
 &= \int_0^1 g_i'(t) dt \\
 &= \int_0^1 Df_i((1-t)x + ty) \cdot (y-x) dt \\
 &= \left(\int_0^1 Df_i((1-t)x + ty) dt \right) \cdot (y-x),
 \end{aligned}$$

where the second equality follows from the fundamental theorem of calculus and the third equality follows from the chain rule. The result follows from collecting all components together. $\square$

The open and closed balls of radius r centered at x_0 are denoted by

$$B_r(x_0) \stackrel{\text{def}}{=} \{x \in X \mid \|x_0 - x\| < r\} \quad \text{and} \quad \overline{B_r(x_0)} \stackrel{\text{def}}{=} \{x \in X \mid \|x_0 - x\| \leq r\}$$

respectively.

In this section, we will be using the ∞ -norm in all examples. Therefore, we explicitly state the following result and leave the proof to the reader.

Corollary 2.2.6 (The mean value inequality). *Consider an open set $U \subset \mathbb{R}^n$. Let $f: U \rightarrow \mathbb{R}^n$ be a C^1 function. Fix a point $x_0 \in U$ and assume $\overline{B_\delta(x_0)} \subset U$ for some $\delta > 0$. Then, for all $x, y \in B_\delta(x_0)$,*

$$\|f(y) - f(x)\|_\infty \leq \left(\sup_{z \in \overline{B_\delta(x_0)}} \|Df(z)\|_\infty \right) \|y - x\|_\infty, \quad (2.6)$$

where $\|Df(\cdot)\|_\infty$ denotes the operator norm.

Observe that since Df is continuous on $\overline{B_\delta(x_0)}$,

$$\sup_{z \in \overline{B_\delta(x_0)}} \|Df(z)\|_\infty$$

is finite.

2.3 Newton's Method

As indicated in the introduction, the goal of this section is to develop techniques by which we can guarantee the existence of zeros of a function. The contraction mapping theorem provides a means of proving the existence of a unique fixed points in a purely topological

setting. The following proposition – the proof is left to the reader – relates fixed points with zeros. For the remainder of this book we use the notation $\tilde{x}$ to denote a zero of a function, that is, $f(\tilde{x}) = 0$.

Proposition 2.3.1. *Let X and Y be vector spaces. Let $U \subset X$ and consider $f: U \rightarrow Y$. Assume that $A: Y \rightarrow X$ is an injective linear map. Let $T: U \rightarrow X$ be defined by*

$$T(x) \stackrel{\text{def}}{=} x - Af(x). \quad (2.7)$$

Then, $T(\tilde{x}) = \tilde{x}$ if and only if $f(\tilde{x}) = 0$.

If $X = Y$ is a finite dimensional vector space, then the assumption that A is injective is equivalent to A being invertible.

Via Proposition 2.3.1 to find a zero of f it is sufficient to find an injective linear map A that makes T a contraction. This leads us to Newton's method and the beginning of the analytic approach of this book.

We make use of the following notation. Given subsets of normed vector spaces $U \subset X$ and $V \subset Y$, the statement $f \in C^r(U, V)$ indicates that $f: U \rightarrow V$, f is r -times differentiable, and the r -th derivative of f is continuous.

Consider $f \in C^1(\mathbb{R}, \mathbb{R})$. Recall that in Newton's method the fixed point map T is given by

$$T(x) \stackrel{\text{def}}{=} x - \frac{1}{f'(x)}f(x)$$

and (as is described in Section 2.1) T is used iteratively to find a fixed point $\tilde{x}$, and hence a zero of f . Observe that if T is continuous at $\tilde{x}$ (a sufficient condition for this is that $f'(\tilde{x}) \neq 0$), then $T(\tilde{x}) = \tilde{x}$ and hence $f(\tilde{x}) = 0$. Thus the problem of proving the existence of a zero of f is essentially reduced to finding and/or identifying whether an initial guess $\bar{x} \in \mathbb{R}$ leads to convergence of Newton's method.

Newton's method and its variants are widely used to find zeros of functions. The contraction mapping theorem provides insight into why this is such a powerful formalism. For simplicity assume that $f \in C^2$ and, more importantly, that $f'(\tilde{x}) \neq 0$. In this case we can compute the derivative

$$T'(\tilde{x}) = 1 - \frac{(f'(\tilde{x}))^2 - f(\tilde{x})f''(\tilde{x})}{(f'(\tilde{x}))^2} = 0. \quad (2.8)$$

This implies that T' is small near the fixed point $\tilde{x}$ of T , hence, as we shall argue, T has a small contraction constant in a neighborhood of $\tilde{x}$.

Indeed, consider the open ball $B_\delta(\tilde{x})$ where $\delta > 0$ is sufficiently small that

$$\kappa := \sup_{\xi \in \overline{B_\delta(\tilde{x})}} |T'(\xi)| < 1.$$

The mean value inequality implies that for any $x, y \in B_\delta(\tilde{x})$,

$$|T(x) - T(y)| \leq \kappa|x - y|,$$

and therefore, T has a contraction constant κ on $B_\delta(\tilde{x})$. Furthermore, κ goes to 0 as δ goes to 0.

The intended take away message from this discussion is that if the derivative is non-zero, then in a sufficiently small neighborhood of a zero of f the associated Newton operator is an extremely strong contraction. In particular, returning to the question concerning the choice of an invertible linear map A , the straightforward but naive interpretation of this example suggests setting $A = A(x) = (f'(x))^{-1}$.

We wish to generalize this discussion to higher dimensions.

Definition 2.3.2. Let $f \in C^1(U, \mathbb{R}^n)$ where $U \subset \mathbb{R}^n$ is an open set. A point $\tilde{x} \in U$ is a *nondegenerate zero* of f if $f(\tilde{x}) = 0$ and $Df(\tilde{x})$ is invertible.

If $f \in C^2(U, \mathbb{R}^n)$ where $U \subset \mathbb{R}^n$ is an open set, then the associated Newton operator is given by

$$T(x) \stackrel{\text{def}}{=} x - (Df(x))^{-1}f(x). \quad (2.9)$$

An argument similar to that just presented demonstrates that if $\tilde{x}$ is a nondegenerate zero, then in a small neighborhood of $\tilde{x}$, T is a contraction mapping with small contraction constant. Again, this suggests the choice of $A(x) = (Df(x))^{-1}$. However, in practice the computational cost of inverting an $N \times N$ matrix is of order N^3 , and therefore for high dimensional problems repeatedly computing the inverse is prohibitively expensive.

Another, and perhaps more substantial reason for not wanting to use $A(x) = (Df(x))^{-1}$ is that for most of this book we are interested in applying these techniques to maps that are defined on infinite dimensional spaces. In this setting, finding an explicit representation of $(Df(x))^{-1}$ is typically not possible. Thus the approach we adopt is as follows. We assume that we are given an initial guess $\bar{x}$ for a zero of f (in practice $\bar{x}$ is obtained numerically) and then use some, typically problem dependent, form of approximation of $(Df(\bar{x}))^{-1}$ to choose A .

As the careful reader may have noted, we have in the previous two paragraphs implicitly made use of a metric on linear maps. To be more explicit, the analogue of (2.8) is

$$DT(\tilde{x}) = I - D(Df(x))^{-1} \Big|_{x=\tilde{x}} f(\tilde{x}) - (Df(\tilde{x}))^{-1}Df(\tilde{x}) = 0$$

from which we concluded that the linear map $DT(z)$ is small if $\|z - \tilde{x}\|$ is small. As in the scalar case, the mean value inequality (Corollary 2.2.6) implies that T has a contraction constant $\kappa < 1$ on $B_\delta(\tilde{x})$ for small enough $\delta > 0$.

2.4 Radii Polynomials in Finite Dimensions

Throughout this book we use the contraction mapping theorem applied to functions of the form $T(x) = x - Af(x)$ to identify zeroes of f . Two key issues remain to be resolved: how does one choose the domain of T , and how does one prove that T is a contraction on this domain?

Consider $f(x) = x^2 - 1$. The associated Newton operator $T: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ is given by $T(x) = x - (2x)^{-1}(x^2 - 1)$. Since T has two distinct fixed points, $T(\pm 1) = \pm 1$, T cannot be a contraction mapping over its entire domain. However, the discussion of the previous section suggests that there exists a neighborhood U^+ of 1 such that $T: U^+ \rightarrow \mathbb{R}$ defined by $T(x) \stackrel{\text{def}}{=} x - \frac{1}{2}(x^2 - 1)$ is a contraction mapping. The theorem below provides a mechanism for rigorously identifying a domain on which T is a contraction mapping.

Theorem 2.4.1. *Consider a map $T \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ and let $\bar{x} \in \mathbb{R}^n$. Let $Y_0 \geq 0$ and $Z: (0, \infty) \rightarrow [0, \infty)$ be a non-negative function satisfying*

$$\|T(\bar{x}) - \bar{x}\| \leq Y_0 \quad (2.10)$$

$$\|DT(c)\| \leq Z(r), \quad \text{for all } c \in \overline{B_r(\bar{x})} \text{ and all } r > 0. \quad (2.11)$$

Define

$$p(r) \stackrel{\text{def}}{=} (Z(r) - 1)r + Y_0. \quad (2.12)$$

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then there exists a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ such that $T(\tilde{x}) = \tilde{x}$. The same result holds when $\mathbb{R}^n$ is replaced by $\mathbb{C}^n$.

Proof. The assumption that $p(r_0) < 0$ is equivalent to $Z(r_0)r_0 \leq Z(r_0)r_0 + Y_0 < r_0$ and hence, using that $r_0 > 0$,

$$Z(r_0) < 1. \quad (2.13)$$

We first show that T maps $\overline{B_{r_0}(\bar{x})}$ into itself. Let $x \in \overline{B_{r_0}(\bar{x})}$. Apply the Mean Value Theorem (see Corollary 2.2.6) leads to

$$\begin{aligned} \|T(x) - \bar{x}\| &\leq \|T(x) - T(\bar{x})\| + \|T(\bar{x}) - \bar{x}\| \\ &\leq \sup_{c \in \overline{B_{r_0}(\bar{x})}} \|DT(c)\| \|x - \bar{x}\| + Y_0 \\ &\leq Z(r_0)r_0 + Y_0 \\ &< r_0, \end{aligned}$$

where the last inequality follows from the assumption that $p(r_0) < 0$. Hence, $T(\overline{B_{r_0}(\bar{x})}) \subset \overline{B_{r_0}(\bar{x})}$.

To see that T is a contraction on $\overline{B_{r_0}(\bar{x})}$, let $a, b \in \overline{B_{r_0}(\bar{x})}$ and apply the mean value inequality to obtain

$$\|T(a) - T(b)\| \leq \sup_{c \in \overline{B_{r_0}(\bar{x})}} \|DT(c)\| \|a - b\| \leq Z(r_0) \|a - b\|.$$

Since $Z(r_0) < 1$, T is a contraction. The contraction mapping theorem yields the existence of a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ such that $T(\tilde{x}) = \tilde{x}$. $\square$

The following corollary of Theorem 2.4.1 provides an efficient means of proving existence of nondegenerate zeroes of nonlinear maps. It is a variant of the well-known Newton-Kantorovich Theorem.

Theorem 2.4.2. [Radii polynomials in finite dimensions] *Consider $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$. Let $\bar{x} \in \mathbb{R}^n$ and $A \in M_n(\mathbb{R})$. Let Y_0 and Z_0 be non-negative constants and $Z_2: (0, \infty) \rightarrow [0, \infty)$ be a non-negative function satisfying*

$$\|Af(\bar{x})\| \leq Y_0 \quad (2.14)$$

$$\|I - ADf(\bar{x})\| \leq Z_0 \quad (2.15)$$

$$\|A[Df(c) - Df(\bar{x})]\| \leq Z_2(r)r, \quad \text{for all } c \in \overline{B_r(\bar{x})} \text{ and all } r > 0. \quad (2.16)$$

Define

$$p(r) \stackrel{\text{def}}{=} Z_2(r)r^2 - (1 - Z_0)r + Y_0. \quad (2.17)$$

If there exists $r_0 > 0$ such that $p(r_0) < 0$, then there exists a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ satisfying $f(\tilde{x}) = 0$ and $Df(\tilde{x})$ is invertible (hence $\tilde{x}$ is a nondegenerate zero).

The same result holds when $\mathbb{R}^n$ is replaced by $\mathbb{C}^n$.

It may appear odd that no constant Z_1 appears in Theorem 2.4.2, however we reserve this symbol for Theorem 7.6.2, the analogous infinite dimensional result.

Definition 2.4.3. Since in practice the bound $Z_2(r)$ is usually chosen to be a polynomial with nonnegative coefficients, the function $p(r)$ is called a *radii polynomial*.

Proof. The assumption that $p(r_0) < 0$ is equivalent to

$$Z(r_0) \stackrel{\text{def}}{=} Z_0 + Z_2(r_0)r_0 < 1 - \frac{Y_0}{r_0} \leq 1. \quad (2.18)$$

Define the Newton-like mapping T by

$$T(x) = x - Af(x).$$

Hence, $\|T(\bar{x}) - \bar{x}\| = \|Af(\bar{x})\| \leq Y_0$, and given $c \in \overline{B_{r_0}(\bar{x})}$,

$$\begin{aligned} \|DT(c)\| &= \|I - ADf(c)\| \\ &\leq \|I - ADf(\bar{x})\| + \|A[Df(\bar{x}) - Df(c)]\| \\ &\leq Z_0 + Z_2(r_0)r_0 && \text{by (2.15) and (2.16)} \end{aligned} \quad (2.19)$$

$$= Z(r_0) < 1 && \text{by (2.18).} \quad (2.20)$$

In particular, $\|I - ADf(\bar{x})\| < 1$ and hence $ADf(\bar{x})$ is invertible (see Exercise 2.7.1). This implies that A is invertible.

Recalling (2.17) and (2.18), note that

$$p(r_0) = Z_2(r_0)r_0^2 - (1 - Z_0)r_0 + Y_0 = (Z_0 + Z_2(r_0)r_0 - 1)r_0 + Y_0 = (Z(r_0) - 1)r_0 + Y_0,$$

which recovers the formulation (2.12). We conclude from Theorem 2.4.1 the existence of a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ such that $T(\tilde{x}) = \tilde{x} - Af(\tilde{x}) = \tilde{x}$. By Proposition 2.3.1 the invertibility of A yields the existence of a unique solution $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ of $f = 0$. Recalling (2.20), we get that for every $c \in \overline{B_{r_0}(\bar{x})}$, $\|DT(c)\| = \|I - ADf(c)\| < 1$. In particular, for $c = \tilde{x} \in \overline{B_{r_0}(\bar{x})}$, $\|I - ADf(\tilde{x})\| < 1$, which implies (by the same argument as above) that $Df(\tilde{x})$ is invertible. We conclude that $\tilde{x}$ is a nondegenerate zero of f . $\square$

Remark 2.4.4. Consider a radii polynomial $p(r) = Z_2(r)r^2 - (1 - Z_0)r + Y_0$ for a function $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ and assume that $Z_2(r)$ is a polynomial with non-negative coefficients, or more generally a non-decreasing function of r . Observe that if there exists $r_0 > 0$ such that $p(r_0) < 0$, then there exists an interval $(r_-, r_+) \subset (0, \infty)$ containing r_0 over which the inequality is satisfied. The maximal such interval is called the *existence interval* for the radii polynomial p and denoted by

$$\text{El}(p).$$

Since $\tilde{x}$ is the unique zero of f in $\overline{B_r(\bar{x})}$ for all $r \in \text{El}(p)$, r_- provides tight bounds on the location of $\tilde{x}$, while r_+ provides information about the domain of isolation of $\tilde{x}$. In particular, if the existence interval for the radii polynomials is nonempty, then one can present an explicit domain $U \subset \mathbb{R}^n$ in which there exists a unique zero of f .

To provide insight into the applicability of Theorem 2.4.2 we consider a few elementary examples.

Example 2.4.5. Consider the simplest nonlinear function $f(x) = x^2 - 2$. The first step in the application of the radii polynomial approach is to choose an initial guess $\bar{x}$ for the zero of f . In a typical application $\bar{x}$ is obtained via a numerical computation. The next step is to choose a matrix $A \in M_1(\mathbb{R}) = \mathbb{R}$. The discussion of Section 2.3 suggests that A be chosen based on $Df(\bar{x})^{-1}$. We leave it general for now. To apply Theorem 2.4.2 we need to determine bounds Y_0 , Z_0 and Z_2 satisfying (2.14), (2.15), and (2.16), respectively.

Consider any Y_0 and Z_0 such that

$$|A(\bar{x}^2 - 2)| =: Y_0, \tag{2.21}$$

$$|1 - ADf(\bar{x})| = |1 - A(2\bar{x})| =: Z_0. \tag{2.22}$$

If $c \in \overline{B_r(\bar{x})} = [\bar{x} - r, \bar{x} + r]$, then

$$|A[Df(c) - Df(\bar{x})]| = |A(2c - 2\bar{x})| = 2|A||c - \bar{x}| \leq 2|A|r = Z_2r,$$

that is $Z_2 \stackrel{\text{def}}{=} 2|A|$. The resulting radii polynomial (2.17) is given by

$$\begin{aligned} p(r) &= Z_2 r^2 - (1 - Z_0)r + Y_0 \\ &= 2|A|r^2 - (1 - |1 - 2\bar{x}A|)r + |A(\bar{x}^2 - 2)|. \end{aligned}$$

Let us now make some explicit choices.

The optimal choice. Let $\bar{x} = \sqrt{2}$ (this is an exact solution) and $A = Df(\bar{x})^{-1} = \frac{1}{2\bar{x}}$ (the exact inverse). In this case, the radii polynomial is

$$p(r) = 2Ar^2 - (1 - |1 - 2\bar{x}A|)r + |A(\bar{x}^2 - 2)| = \frac{\sqrt{2}}{2}r^2 - r.$$

The existence interval for the radii polynomial is $\text{El}(p) = (r_-, r_+) = (0, \sqrt{2})$. Since $r_- = 0$ we know that $\tilde{x} = \sqrt{2}$ is the exact solution. Since $r_+ = \sqrt{2}$ we know that $\tilde{x} = \sqrt{2}$ is the unique solution in the domain $(0, 2\sqrt{2})$.

Choices based on approximations. In applications we cannot expect to have exact values. With this in mind choose $\bar{x} = 1.3$. Observe that $Df(\bar{x})^{-1} = (2\bar{x})^{-1} \approx 0.384615$. For simplicity choose $A = 0.38$.

We need to determine bounds Y_0 , Z_0 and Z_2 . Again, in practice one cannot expect to obtain optimal bounds. Observe that given our choice of $\bar{x}$ and A

$$|Af(\bar{x})| = |0.38(1.3^2 - 2)| = 0.1178$$

and thus by (2.21) we can choose $Y_0 \stackrel{\text{def}}{=} 0.12$. Using (2.22) we choose $Z_0 = 0.012$ and set $Z_2 = 0.76$. The associated radii polynomial is given by

$$p(r) = Z_2 r^2 - (1 - Z_0)r + Y_0 = 0.76r^2 - 0.988r + 0.12$$

and its graph is shown in Figure 2.1.

The quadratic formula implies that $p(r) < 0$ for all $r \in I = [0.136, 1.164]$, and therefore by Theorem 2.4.2 and Remark 2.4.4 we can conclude that $\tilde{x} \in \overline{B_{0.136}(1.3)} = [1.164, 1.436]$. Given that the actual root contained in this interval is $\sqrt{2} \approx 1.414$, the relative error for this bound is

$$\frac{1.414 - 1.164}{1.414} \approx 0.1768.$$

We can also conclude that there is a unique root in the interval $\tilde{x} \in \overline{B_{1.164}(1.3)} = [0.136, 2.464]$.

Note that for different values of $r > 0$, the value of $p(r)$ leads to different scenarios. If the radius r is too small and $p(r) \geq 0$, then the method based on the radii polynomials cannot conclude about the existence of a unique solution $x^* \in \overline{B_r(\bar{x})}$. For instance considering $r = 0.1 \notin I = [0.136, 1.164]$, the ball is too small to enclose the true solution (see the left picture of Figure 2.2). By choosing $r \in I$ (for instance $r = 0.15$), then $p(r) < 0$, which

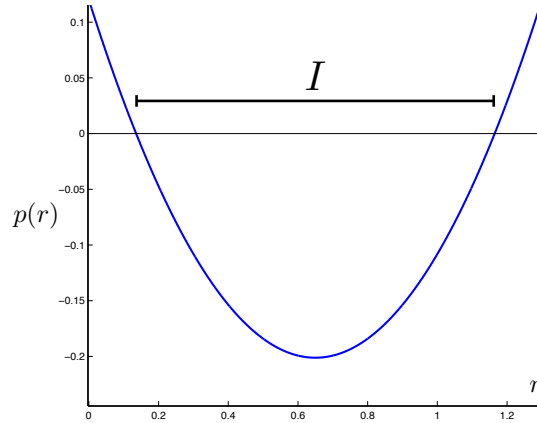


Figure 2.1: The radii polynomial $p(r) = 0.76r^2 - 0.988r + 0.12$ of Example 2.4.5 and $I = [0.136, 1.164]$, where the radii polynomial is strictly negative.

implies that the ball $\overline{B_r(\bar{x})}$ contains a unique solution (see the picture in the center of Figure 2.2). Choosing the radius $r \notin I$ too large leads to $p(r) \geq 0$ and the radii polynomial approach fails, but now for different reasons than before. First, there could be more than one solution in the ball $\overline{B_r(\bar{x})}$. For instance for $r = 3$, the ball $\overline{B_r(\bar{x})} = \overline{B_3(1.3)}$ contains the two solutions $x^* = \sqrt{2}$ and $x_1^* = -\sqrt{2}$ of $f(x) = 0$ (see the right picture of Figure 2.2). Second, the Newton-like map could simply not be a contraction on the $\overline{B_r(\bar{x})}$ since the radius is too large (keep in mind that Newton's method is a contraction only on a small enough neighborhood of the solution).

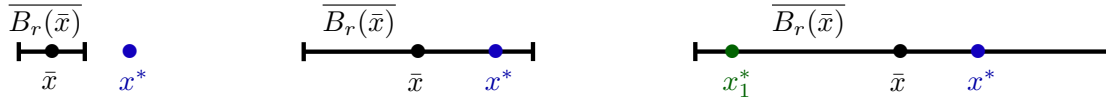


Figure 2.2: For different values of $r > 0$, we have different scenarios.

Example 2.4.5 demonstrates how the radii polynomial can be employed. It also makes clear that the choice of A plays a central role in the values of Y_0 , Z_0 and Z_2 , which in turn determines whether the resulting radii polynomial inequality can be satisfied. With this in mind we seek to gain some insight into how broad a range of A is possible.

Example 2.4.6. As a slight generalization of Example 2.4.5, let $f(x) = x^2 - \lambda$ and consider $\bar{x} > 0$. Using Newton's method as a guide we set $A = (f'(\bar{x}))^{-1} = (2\bar{x})^{-1}$. By (2.15) we can choose $Z_0 = 0$. Using (2.14) we choose $Y_0 = |Af(\bar{x})|$. Applying the same analysis as in Example 2.4.5 set $Z_2 = 2|A| = \frac{1}{\bar{x}}$. With these choices the radii polynomial takes the form

$$p(r) = \frac{1}{\bar{x}}r^2 - r + \frac{|\bar{x}^2 - \lambda|}{2\bar{x}}$$

and therefore the endpoints of the associate existence interval for $\lambda \geq 0$ are given by

$$r_{\pm} = \frac{\bar{x} \pm \sqrt{\bar{x}^2 - 2|\bar{x}^2 - \lambda|}}{2}.$$

This in turn tells us that if

$$\bar{x} \in \left(\sqrt{\frac{2\lambda}{3}}, \sqrt{2\lambda} \right),$$

then $\bar{x}^2 - 2|\bar{x}^2 - \lambda| > 0$ and therefore the radii polynomials will provide a positive answer to the question of existence of a zero of f .

Example 2.4.7. In the previous examples $Z_2(r)$ is chosen to be a constant. This has to do with the fact that the nonlinearity is quadratic. Consider the problem of finding the zeros of

$$f(x) = x^3 + 2x^2 - 5x - 10.$$

The exact answers are $\tilde{x} = -2, \pm\sqrt{5}$. Consider $\bar{x} = 2.24$.

Simple calculations show that $f(\bar{x}) = 0.074624$ and $f'(\bar{x}) = 19.0128$.

From the discussion of the optimal choice in Example 2.4.5, we set $A = (f'(\bar{x}))^{-1} > 0$. Using (2.14) and (2.15) we choose

$$\begin{aligned} Y_0 &:= 0.004 > \frac{0.074624}{19.0128} = |Af(\bar{x})| \\ Z_0 &:= 0 = |1 - Af'(\bar{x})|. \end{aligned}$$

To determine $Z_2(r)$ it is useful to adopt a change of variables. Given $c \in \overline{B_r(\bar{x})}$, set $b := c - \bar{x} \in \overline{B_r(0)}$. Hence,

$$\begin{aligned} |A[f'(c) - f'(\bar{x})]| &= |A[f'(\bar{x} + b) - f'(\bar{x})]| \\ &= |A[3(\bar{x} + b)^2 + 4(\bar{x} + b) - 5 - (3\bar{x}^2 + 4\bar{x} - 5)]| \\ &= |A[3(2\bar{x}b + b^2) + 4b]| \\ &\leq A(6\bar{x} + 4 + 3r)r, \end{aligned}$$

where the last inequality arises since $b \in \overline{B_r(0)}$, that is $|b| \leq r$. Thus we can choose

$$Z_2(r) := A(6\bar{x} + 4 + 3r) = \frac{1}{19.0128}(17.44 + 3r).$$

The resulting radii polynomial is now cubic, and is given by

$$p(r) = \frac{3}{19.0128}r^3 + \frac{17.44}{19.0128}r^2 - r + 0.004$$

for which $[0.005, 0.93]$ is contained in the associated existence interval.

Example 2.4.8. Consider the use of Theorem 2.4.2 to the problem of verifying zeros of $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$f(x) = \begin{pmatrix} 4x_1^2 + x_2 - \lambda \\ x_1 + x_2^2 - 1 \end{pmatrix} \quad (2.23)$$

where $x = (x_1, x_2)$ and $\lambda \in \mathbb{R}$ is a parameter. This corresponds to finding the intersecting points of the parabolas $4x_1^2 + x_2 = \lambda$ and $x_1 + x_2^2 = 1$ (see Figure 2.3 for the case $\lambda = 3$).

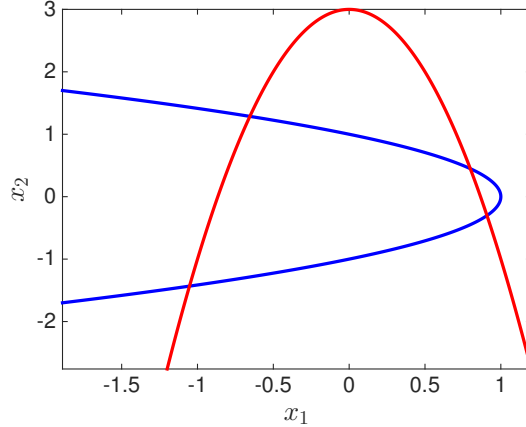


Figure 2.3: Intersection of the parabolas $4x_1^2 + x_2 = 3$ (in red) and $x_1 + x_2^2 = 1$ (in blue).

To construct a radii polynomial we need to determine bounds Y_0 , Z_0 and Z_2 satisfying (2.14), (2.15), and (2.16), respectively.

Because (2.23) is relatively simple, given an initial guess $\bar{x}$ we can explicitly compute

$$Df(\bar{x}) = \begin{pmatrix} 8\bar{x}_1 & 1 \\ 1 & 2\bar{x}_2 \end{pmatrix} \quad \text{and} \quad Df(\bar{x})^{-1} = \frac{1}{16\bar{x}_1\bar{x}_2 - 1} \begin{pmatrix} 2\bar{x}_2 & -1 \\ -1 & 8\bar{x}_1 \end{pmatrix},$$

assuming of course that $16\bar{x}_1\bar{x}_2 - 1 \neq 0$. The optimal choice of Example 2.4.5 suggests setting $A \stackrel{\text{def}}{=} Df(\bar{x})^{-1}$.

With this choice of A we can set $Y_0 := \|Df(\bar{x})^{-1}f(\bar{x})\|$. Furthermore, observe that $I - ADf(\bar{x}) = 0$, which allows us to set $Z_0 = 0$.

To determine Z_2 it is useful to use a change of variables. Given $c \in \overline{B_r(\bar{x})}$, set $b := c - \bar{x} \in \overline{B_r(0)}$. Then

$$\begin{aligned} (Df(c) - Df(\bar{x})) &= (Df(\bar{x} + b) - Df(\bar{x})) \\ &= \left[\begin{pmatrix} 8\bar{x}_1 + 8b_1 & 1 \\ 1 & 2\bar{x}_2 + 2b_2 \end{pmatrix} - \begin{pmatrix} 8\bar{x}_1 & 1 \\ 1 & 2\bar{x}_2 \end{pmatrix} \right] \\ &= \begin{pmatrix} 8b_1 & 0 \\ 0 & 2b_2 \end{pmatrix} \end{aligned}$$

and hence

$$A(Df(c) - Df(\bar{x})) = \frac{1}{16\bar{x}_1\bar{x}_2 - 1} \begin{pmatrix} 16\bar{x}_2b_1 & -2b_2 \\ -8b_1 & 16\bar{x}_1b_2 \end{pmatrix}.$$

To make a choice for Z_2 requires that we obtain an explicit bound for $\|A(Df(c) - Df(\bar{x}))\|$. This requires that we make an explicit choice of norm. As discussed in Remark 2.2.4, since we know the entries of the matrix $A(Df(c) - Df(\bar{x}))$ it is straightforward to compute the 1-norm or the sup-norm.

Choosing the sup-norm, by (2.5)

$$\begin{aligned} \|A(Df(c) - Df(\bar{x}))\|_\infty &= \max \left\{ \frac{16|\bar{x}_2b_1| + 2|b_2|}{|16\bar{x}_1\bar{x}_2 - 1|}, \frac{8|b_1| + 16|\bar{x}_1b_2|}{|16\bar{x}_1\bar{x}_2 - 1|} \right\} \\ &\leq \max \left\{ \frac{16|\bar{x}_2| + 2}{|16\bar{x}_1\bar{x}_2 - 1|}, \frac{8 + 16|\bar{x}_1|}{|16\bar{x}_1\bar{x}_2 - 1|} \right\} r \end{aligned}$$

where the inequality arises from the fact that $\|b\|_\infty = \max\{|b_1|, |b_2|\} \leq r$. Thus, we can set

$$Z_2 := \max \left\{ \frac{16|\bar{x}_2| + 2}{|16\bar{x}_1\bar{x}_2 - 1|}, \frac{8 + 16|\bar{x}_1|}{|16\bar{x}_1\bar{x}_2 - 1|} \right\}.$$

The resulting radii polynomial (2.17) is

$$p(r) \stackrel{\text{def}}{=} \max \left\{ \frac{16|\bar{x}_2| + 2}{|16\bar{x}_1\bar{x}_2 - 1|}, \frac{8 + 16|\bar{x}_1|}{|16\bar{x}_1\bar{x}_2 - 1|} \right\} r^2 - r + \|Df(\bar{x})^{-1}f(\bar{x})\|_\infty. \quad (2.24)$$

Observe that we have derived a radii polynomial that is valid independently of the approximate solution $\bar{x} \in \mathbb{R}^2$. Thus, our expectation is that if we are given any reasonable approximation of a zero of (2.23), then there will exist $r_0 > 0$ such that $p(r_0) < 0$.

To test our expectation, we set $\lambda = 3$ and applied a numerical scheme based on Newton's method to find $\bar{x} \in \mathbb{R}^2$ such that $\|f(\bar{x})\| < \text{tol}$ where we fixed the tolerance $\text{tol} = 10^{-15}$. This resulted in four approximate solutions

$$\begin{aligned} \bar{x}^{(1)} &= \begin{pmatrix} -0.6545436118927946 \\ 1.286290640521338 \end{pmatrix}, & \bar{x}^{(2)} &= \begin{pmatrix} 0.7986333753610425 \\ 0.4487389270377125 \end{pmatrix}, \\ \bar{x}^{(3)} &= \begin{pmatrix} 0.9086121587039679 \\ -0.3023042197787387 \end{pmatrix}, & \bar{x}^{(4)} &= \begin{pmatrix} -1.052701922172216 \\ -1.432725347780312 \end{pmatrix}. \end{aligned} \quad (2.25)$$

For each choice of approximate solution we obtain a radii polynomial

$$p^{(j)}(r) \stackrel{\text{def}}{=} \max \left\{ \frac{16|\bar{x}_2^{(j)}| + 2}{|16\bar{x}_1^{(j)}\bar{x}_2^{(j)} - 1|}, \frac{8 + 16|\bar{x}_1^{(j)}|}{|16\bar{x}_1^{(j)}\bar{x}_2^{(j)} - 1|} \right\} r^2 - r + \|Df(\bar{x}^{(j)})^{-1}f(\bar{x}^{(j)})\|_\infty.$$

For $i = 1, 2, 3, 4$ one can check that the following intervals

$$\begin{aligned} I^{(1)} &= [2.5 \times 10^{-16}, 0.640] \subset \text{El}(p^{(1)}) \\ I^{(2)} &= [6 \times 10^{-16}, 0.227] \subset \text{El}(p^{(2)}) \\ I^{(3)} &= [1.7 \times 10^{-16}, 0.239] \subset \text{El}(p^{(3)}) \\ I^{(4)} &= [2.2 \times 10^{-16}, 0.927] \subset \text{El}(p^{(4)}) \end{aligned} \tag{2.26}$$

are subset of the existence interval for the associated radii polynomial (2.24). Figure 2.4 shows $\overline{B_{r_+^{(i)}}(\bar{x}^{(i)})}$ using the sup-norm where $r_+^{(i)}$ is the upper bound of $I^{(i)}$.

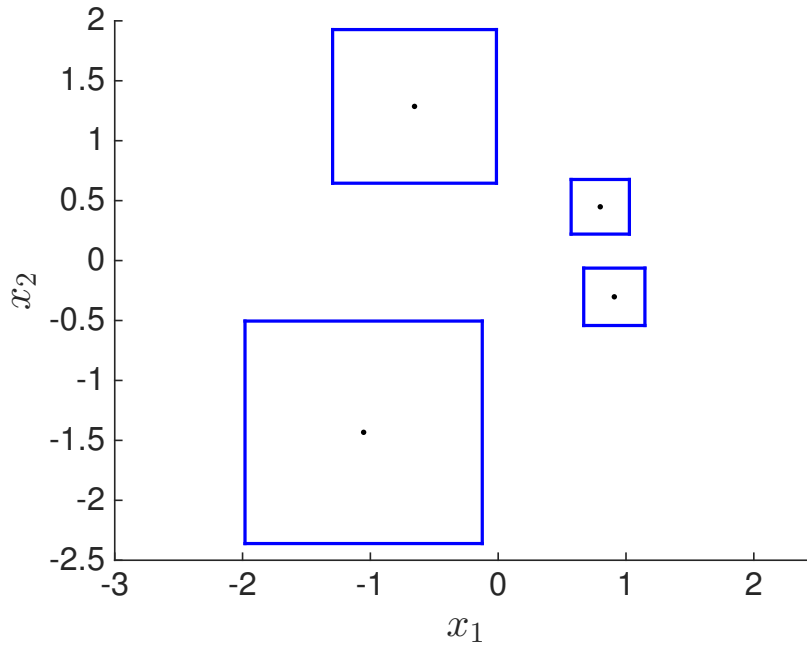


Figure 2.4: Largest existence and uniqueness enclosures for each equilibrium of (2.23) for $\lambda = 3$. For each $i = 1, 2, 3, 4$, the radius around $\bar{x}^{(i)}$ is the largest value of $I^{(i)}$. The quantities $\bar{x}^{(i)}$ and $I^{(i)}$ are found in (2.25) and (2.26) respectively. The smallest enclosure is too small to represent graphically, which implies that the dots representing $\bar{x}^{(i)}$ also represent the true equilibria $\tilde{x}^{(i)}$.

In the previous examples we had the option of setting $A := Df(\bar{x})^{-1}$, which allows us to set $Z_0 = 0$. For higher dimensional problems this is not feasible, but as the following example demonstrates, having the exact inverse is not necessary.

Example 2.4.9. The Lorenz equations

$$\begin{aligned}\dot{x}_1 &= \sigma(x_2 - x_1) \\ \dot{x}_2 &= \rho x_1 - x_2 - x_1 x_3 \\ \dot{x}_3 &= -\beta x_3 + x_1 x_2\end{aligned}\tag{2.27}$$

provides one of the canonical examples of chaotic dynamics. We will use it as a simple benchmark for the theorems and techniques developed in this book. Typically, the first question one asks of a system of differential equations is: does it exhibit equilibria? Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by the right-hand side of (2.27). Then, equilibria for the Lorenz equations are equivalent to zeros of f . Since f is a quadratic polynomial it is not surprising that exact solutions for the equilibria can be derived. In particular, for any $\beta > 0$ and $\rho > 1$, the set of equilibria of (2.27) is given by

$$\left\{ (0, 0, 0), \left(\pm \sqrt{\beta(\rho - 1)}, \pm \sqrt{\beta(\rho - 1)}, \rho - 1 \right) \right\}.\tag{2.28}$$

With the answer in hand we can test the efficacy of using radii polynomials to identify the existence of an equilibrium from a numerical approximation.

At the classical parameter values [23], $\sigma = 10$, $\rho = 28$ and $\beta = 8/3$, $(\sqrt{72}, \sqrt{72}, 27)$ is an equilibrium solution that is approximated by

$$\bar{x} \stackrel{\text{def}}{=} \begin{pmatrix} 8.4853 \\ 8.4853 \\ 27 \end{pmatrix}.\tag{2.29}$$

Observe that

$$Df(\bar{x}) = \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - \bar{x}_3 & -1 & -\bar{x}_1 \\ \bar{x}_2 & \bar{x}_1 & -\beta \end{pmatrix}.$$

Rather than looking for an explicit expression for $Df(\bar{x})^{-1}$ we compute a numerical inverse and set

$$A \stackrel{\text{def}}{=} \begin{pmatrix} -0.051851843722001 & -0.018518437220007 & 0.058925435753597 \\ 0.048148156277999 & -0.018518437220007 & 0.058925435753597 \\ -0.011785087150719 & -0.117850871507195 & 0 \end{pmatrix}.$$

To obtain explicit values for Y_0 , Z_0 and Z_2 we need to choose a norm. For computational efficacy we choose the sup-norm.

We can compute the following bounds

$$Y_0 = \|Af(\bar{x})\|_\infty\tag{2.30}$$

and

$$Z_0 = \|I - ADf(\bar{x})\|_\infty.\tag{2.31}$$

To obtain Z_2 we use the standard change of variables, for $c \in \overline{B_r(\bar{x})}$ set $b \stackrel{\text{def}}{=} c - \bar{x} \in \overline{B_r(0)}$, and investigate

$$\begin{aligned} Df(c) - Df(\bar{x}) &= Df(b + \bar{x}) - Df(\bar{x}) \\ &= \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - (\bar{x}_3 + b_3) & -1 & -(\bar{x}_1 + b_1) \\ \bar{x}_2 + b_2 & \bar{x}_1 + b_1 & -\beta \end{pmatrix} - \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - \bar{x}_3 & -1 & -\bar{x}_1 \\ \bar{x}_2 & \bar{x}_1 & -\beta \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ -b_3 & 0 & -b_1 \\ b_2 & b_1 & 0 \end{pmatrix}. \end{aligned}$$

Thus, denoting $A = [a_{i,j}]_{i,j=1}^3$ and using (2.5)

$$\begin{aligned} \|A[Df(c) - Df(\bar{x})]\|_\infty &= \max_{i=1,2,3} \max_{\|b\|_\infty \leq r} \{|a_{i,2}| |b_3| + |a_{i,3}| |b_2| + |a_{i,3}| |b_1| + |a_{i,2}| |b_1|\} \\ &\leq \max_{i=1,2,3} \{2(|a_{i,2}| + |a_{i,3}|)\} r. \end{aligned}$$

Hence we set

$$Z_2 = \max_{i=1,2,3} \{2(|a_{i,2}| + |a_{i,3}|)\} = 0.23570174301439.$$

Applying Theorem 2.4.2 to the resulting radii polynomial

$$p(r) = Z_2 r^2 - (1 - Z_0) r + Y_0$$

allows us to conclude that there is a unique equilibrium to (2.27) in the ball $\overline{B_r(\bar{x})}$ for any

$$r \in [1.8626 \times 10^{-5}, 4.2426] \subset \text{El}(p).$$

The previous examples involve low dimensional systems. One of the advantages of the computer is that we can study higher dimensional systems. A particularly simple means of generating arbitrarily high dimensional systems is to consider approximations of partial differential equations. With this in mind we introduce the following notation.

Definition 2.4.10. Let $x, y \in \mathbb{R}^{2N+1}$ where $x = (x_{-N}, \dots, x_N)$. The *discrete convolution* $x * y \in \mathbb{R}^{2N+1}$ is defined component-wise by

$$(x * y)_n \stackrel{\text{def}}{=} \sum_{\substack{k=-N \\ |n-k| \leq N}}^N x_{n-k} y_k.$$

We leave proof of the following properties of convolution as an exercise.

Proposition 2.4.11. For all $x, y, z \in \mathbb{R}^{2N+1}$ and any scalar $a \in \mathbb{R}$,

- (i) (commutativity) $x * y = y * x$,
- (ii) (distributivity) $x * (y + z) = x * y + x * z$,
- (iii) (scalar multiplication) $(ax) * y = a(x * y) = x * (ay)$.

Differentiating convolution with respect to the components of the vectors is straightforward, but we record it for convenience sake.

Proposition 2.4.12. *Let $x, y \in \mathbb{R}^{2N+1}$ and consider $f: \mathbb{R}^{2(2N+1)} \rightarrow \mathbb{R}^{2N+1}$ defined by $f(x, y) = x * y$. Then,*

$$\frac{\partial f_n}{\partial x_l}(x, y) = \begin{cases} y_{n-l} & \text{if } |n-l| \leq N \\ 0 & \text{otherwise.} \end{cases} \quad \text{and} \quad \frac{\partial f_n}{\partial y_l}(x, y) = \begin{cases} x_{n-l} & \text{if } |n-l| \leq N \\ 0 & \text{otherwise.} \end{cases}$$

Proposition 2.4.13. *Let $x \in \mathbb{R}^{2N+1}$ and consider $f: \mathbb{R}^{2(2N+1)} \rightarrow \mathbb{R}^{2N+1}$ defined by $f(x) = x * x$. Then,*

$$\frac{\partial f_n}{\partial x_l}(x) = \begin{cases} 2x_{n-l} & \text{if } |n-l| \leq N \\ 0 & \text{otherwise.} \end{cases}$$

The following result indicates that it is natural to use the 1-norm $\|\cdot\|_1$ when working with convolution, since in this case it satisfies a fundamental property of a Banach algebra (see Section 7.4).

Proposition 2.4.14. *If $x, y \in \mathbb{R}^{2N+1}$, then*

$$\|x * y\|_1 \leq \|x\|_1 \|y\|_1. \quad (2.32)$$

Proof. The result follows from a direction computation,

$$\begin{aligned} \|x * y\|_1 &= \sum_{n=-N}^N \left| \sum_{\substack{k=-N \\ |n-k| \leq N}}^N x_{n-k} y_k \right| \\ &\leq \sum_{n=-N}^N \sum_{\substack{k=-N \\ |n-k| \leq N}}^N |x_{n-k}| |y_k| \\ &= \sum_{k=-N}^N \sum_{\substack{n=-N \\ |n-k| \leq N}}^N |x_{n-k}| |y_k| \\ &= \sum_{k=-N}^N |y_k| \left(\sum_{\substack{n=-N \\ |n-k| \leq N}}^N |x_{n-k}| \right) \\ &\leq \|x\|_1 \|y\|_1. \end{aligned}$$

□

As suggested above our use of discrete convolution arises from approximations of non-linear partial differential equations and as is made clear in later sections this often imposes extra structure on the vectors $x \in \mathbb{R}^{2N+1}$. This in turn impacts the manner in which we write the convolution operator. For example if $x_k = x_{-k}$, then it is sufficient to work with the nonnegative components of x . This leads to the following definition of the *even discrete convolution* which is defined component-wise by

$$(x *_e y)_n \stackrel{\text{def}}{=} \sum_{\substack{k_1+k_2=n \\ k_i=-N,\dots,N}} x_{|k_1|} y_{|k_2|}, \quad n = 0, \dots, N \quad (2.33)$$

where $x = (x_0, \dots, x_N) \in \mathbb{R}^{N+1}$.

Example 2.4.15. As described in greater detail in Section 13.1, the following system arises from a finite dimensional approximation of a partial differential equation known as Fisher's equation,

$$\dot{x}_k = f_k(x, \lambda) \stackrel{\text{def}}{=} (\lambda - k^2)x_k - \lambda \sum_{\substack{k_1+k_2=k \\ k_i=-N,\dots,N}} x_{|k_1|} x_{|k_2|}, \quad k = 0, \dots, N,$$

where λ is a parameter. Setting K to be the diagonal matrix with diagonal entries k^2 allows us to write this system of equations in a more compact form

$$\dot{x} = f(x, \lambda) \stackrel{\text{def}}{=} (\lambda I - K)x - \lambda x *_e x, \quad x \in \mathbb{R}^{N+1}. \quad (2.34)$$

For instance, $N = 1$ leads to the two-dimensional ODE

$$\dot{x} = f(x, \lambda) = \begin{pmatrix} \lambda x_0 - \lambda(x_0^2 + 2x_1^2) \\ (\lambda - 1)x_1 - \lambda(2x_0x_1) \end{pmatrix},$$

$N = 2$ leads to the three-dimensional ODE

$$\dot{x} = f(x, \lambda) = \begin{pmatrix} \lambda x_0 - \lambda(x_0^2 + 2x_1^2 + 2x_2^2) \\ (\lambda - 1)x_1 - \lambda(2x_0x_1 + 2x_1x_2) \\ (\lambda - 4)x_2 - \lambda(2x_0x_2 + x_1^2) \end{pmatrix},$$

and $N = 3$ leads to the four-dimensional ODE

$$\dot{x} = f(x, \lambda) = \begin{pmatrix} \lambda x_0 - \lambda(x_0^2 + 2x_1^2 + 2x_2^2 + 2x_3^2) \\ (\lambda - 1)x_1 - \lambda(2x_0x_1 + 2x_1x_2 + 2x_2x_3) \\ (\lambda - 4)x_2 - \lambda(2x_0x_2 + x_1^2 + 2x_1x_3) \\ (\lambda - 9)x_3 - \lambda(2x_0x_3 + 2x_1x_2) \end{pmatrix}.$$

Let us now develop the radii polynomial approach to the general case, i.e. for fixed $N \in \mathbb{N}$. Assume $\bar{x} \in \mathbb{R}^{N+1}$ satisfies $f(\bar{x}) \approx 0$. Observe that

$$\begin{aligned} \frac{\partial f_k}{\partial x_0}(\bar{x}) &= \lambda - k^2 - 2\lambda x_k \\ \frac{\partial f_k}{\partial x_l}(\bar{x}) &= \lambda - k^2 - 2\lambda \begin{cases} x_{|k-l|} + x_{k+l} & \text{if } k+l \leq N \\ x_{|k-l|} & \text{otherwise.} \end{cases} \end{aligned}$$

Therefore, we have an explicit representation for $Df(\bar{x})$ and can numerically determine $Df(\bar{x})^{-1}$. Let $A \in M_{N+1}(\mathbb{R})$ such that $A \approx Df(\bar{x})^{-1}$. We can compute the following bounds

$$\begin{aligned} Y_0 &= \|Af(\bar{x})\|_1 \\ Z_0 &= \|I - ADf(\bar{x})\|_1. \end{aligned}$$

In preparation for obtaining Z_2 , for $c \in \overline{B_r(\bar{x})}$ and $h \in \overline{B_1(0)} \subset \mathbb{R}^{N+1}$, observe that

$$\begin{aligned} ([Df(c) - Df(\bar{x})]h)_k &= (-k^2 + \lambda) h_k - 2\lambda \left(\sum_{l=1}^N c_{|k-l|} h_l + \sum_{l=0}^{N-k} c_{k+l} h_l \right) \\ &\quad - (-k^2 + \lambda) h_k + 2\lambda \left(\sum_{l=1}^N \bar{x}_{|k-l|} h_l + \sum_{l=0}^{N-k} \bar{x}_{k+l} h_l \right) \\ &= -2\lambda (c *_e h - \bar{x} *_e h)_k \\ &= -2\lambda ((c - \bar{x}) *_e h)_k. \end{aligned}$$

As a consequence

$$\begin{aligned} \|A[Df(c) - Df(\bar{x})]\|_1 &= \sup_{\|h\|=1} \|A[Df(c) - Df(\bar{x})]h\|_1 \\ &\leq 2|\lambda| \sup_{\|h\|=1} \|A(c - \bar{x}) *_e h\|_1 \\ &\leq 2|\lambda| \sup_{\|h\|=1} \|A\|_1 \|(c - \bar{x}) *_e h\|_1 \\ &\leq 2|\lambda| \sup_{\|h\|=1} \|A\|_1 \|c - \bar{x}\|_1 \|h\|_1 \\ &\leq 2|\lambda| \|A\|_1 r \end{aligned}$$

where the second to the last inequality follows from (2.55). Therefore, we set

$$Z_2 = 2|\lambda| \|A\|_1,$$

and obtain the radii polynomial

$$p(r) = Z_2 r^2 - (1 - Z_0)r + Y_0.$$

2.5 Localized Radii Polynomials in Finite Dimensions

In all the examples discussed in the previous section we considered nonlinear functions whose domain is the entire space $\mathbb{R}^n$. The following reformulations of Theorem 2.4.1 and Theorem 2.4.2 (the proofs are left to the reader) show it is sufficient to assume that f is defined on an open set $U \subset \mathbb{R}^n$.

Theorem 2.5.1. [Localized radii polynomials in finite dimensions: version 1]
Consider a C^1 map $f: U \rightarrow \mathbb{R}^n$ where $U \subset \mathbb{R}^n$ is open. Let $\bar{x} \in U$ and $A \in M_n(\mathbb{R})$. Let $r_ > 0$ be such that $\overline{B_{r_*}(\bar{x})} \subset U$. Let Y_0 and Z be non-negative constants satisfying*

$$\|Af(\bar{x})\| \leq Y_0 \quad (2.35)$$

$$\sup_{c \in \overline{B_{r_*}(\bar{x})}} \|I - ADf(c)\| \leq Z. \quad (2.36)$$

Define the radii polynomial

$$p(r) \stackrel{\text{def}}{=} (Z - 1)r + Y_0. \quad (2.37)$$

If there exists $r_0 \in (0, r_]$ such that $p(r_0) < 0$, then the matrix A is invertible and there exists a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ satisfying $f(\tilde{x}) = 0$.*

Theorem 2.5.2. [Localized radii polynomials in finite dimensions: version 2]
Consider a C^1 map $f: U \rightarrow \mathbb{R}^n$ where $U \subset \mathbb{R}^n$ is open. Let $\bar{x} \in U$ and $A \in M_n(\mathbb{R})$. Let $r_ > 0$ be such that $\overline{B_{r_*}(\bar{x})} \subset U$. Let Y_0 , Z_0 , and $Z_2 : (0, r_*] \rightarrow [0, \infty)$ be such that*

$$\|Af(\bar{x})\| \leq Y_0 \quad (2.38)$$

$$\|I - ADf(\bar{x})\| \leq Z_0 \quad (2.39)$$

$$\|A[Df(c) - Df(\bar{x})]\| \leq Z_2(r)r, \quad \text{for all } c \in \overline{B_r(\bar{x})} \text{ and all } r \in (0, r_*]. \quad (2.40)$$

Define the radii polynomial

$$p(r) \stackrel{\text{def}}{=} Z_2(r)r^2 - (1 - Z_0)r + Y_0. \quad (2.41)$$

If there exists $r_0 \in (0, r_]$ such that $p(r_0) < 0$, then the matrix A is invertible and there exists a unique $\tilde{x} \in \overline{B_{r_0}(\bar{x})}$ satisfying $f(\tilde{x}) = 0$.*

The choice of r_* has a certain level of arbitrariness. In particular, it is not necessary to choose r_* close to the distance of $\bar{x}$ to the boundary of U . In practice we usually choose a rather loose a priori bound r_* on r . After checking that $p(r_0) < 0$ and $r_0 \leq r_*$ we can then invoke Theorem 2.5.1 or Theorem 2.5.2 to conclude that a (unique) zero $\tilde{x}$ of f lies within distance r_0 of $\bar{x}$. One may subsequently vary r_* to optimise the existence interval, which in this case is given by

$$\text{El}(p) = \{r \mid p(r) < 0\} \cap (0, r_*]. \quad (2.42)$$

A smaller r_* will allow a sharper bound Z_2 , but such a more restrictive choice for r_* may nevertheless shrink the existence interval (2.42). When r_* is chosen very small, it can happen that the existence interval is empty even though there is an $r_0 > 0$ such that $p(r_0) < 0$ (but $r_0 > r_*$). This suggests that the choice for r_* needs to be increased.

Remark 2.5.3. In the ∞ -norm it is straightforward to evaluate (with interval arithmetic) the left hand sides of (2.38) and (2.39) to obtain the bounds Y_0 and Z_0 . This holds for an arbitrary function f . Furthermore, the following second derivative estimate provides (via the mean value inequality) a general constant bound $Z_2 = Z_2(r_*)$ (independent of r) satisfying (2.40) in the case of the ∞ -norm:

$$Z_2(r_*) = \sup_{b \in B_{r_*}(\bar{x})} \left(\max_{1 \leq i \leq n} \sum_{1 \leq k, m \leq n} \left| \sum_{1 \leq j \leq n} A_{ij} \frac{\partial^2 f_j}{\partial x_m \partial x_k}(b) \right| \right). \quad (2.43)$$

For any fixed r_* this bound can be computed directly using interval arithmetic, when expressions for the second derivatives of f are available. A proof that the bound (2.43) satisfies (2.40) is left as an exercise (see Exercise 2.7.4). The next example makes use of this bound.

Example 2.5.4. A canonical example from celestial mechanics is the restricted three body problem which arises from Newton's law of gravity applied to three bodies where the mass of one body is assumed to be infinitesimally small. We discuss this problem in greater detail in later chapters, for the moment we remark that the *libration* points arise as critical points of the function

$$V(x) = \frac{x^2}{2} \pm \frac{\mu}{x-1+\mu} \pm \frac{1-\mu}{x+\mu}$$

where the signs are chosen such that each term in the expression is positive and (via a renormalization) the masses of the two bodies with mass are $1-\mu > 0$ and $\mu > 0$. For example, for the restricted three body system consisting of the earth, the moon, and a satellite $\mu \approx (1.0123)^{-1}$.

The critical points of V are the zeros of

$$f(x) := V'(x) = x \pm \frac{-\mu}{(x-1+\mu)^2} \pm \frac{\mu-1}{(x+\mu)^2}.$$

Assume that $\mu = (1.0123)^{-1}$ and $-\mu < x < 1-\mu$. In this case, $V(x) = \frac{x^2}{2} - \frac{\mu}{x-1+\mu} + \frac{1-\mu}{x+\mu}$ and therefore

$$f(x) = x + \frac{\mu}{(x-1+\mu)^2} + \frac{\mu-1}{(x+\mu)^2}$$

and clearly the domain of f is not the entire real line (see the plot of f on Figure 2.5).

Consider $\bar{x} = -0.8369153095697018$ obtained using Newton's method. Choosing $r_* = 0.01$ guarantees that $\overline{B_{r_*}(\bar{x})} \subset (-\mu, 1-\mu)$. We set $A = f'(\bar{x})^{-1}$ and choose $Y_0 = 4 \times 10^{-17} \geq$

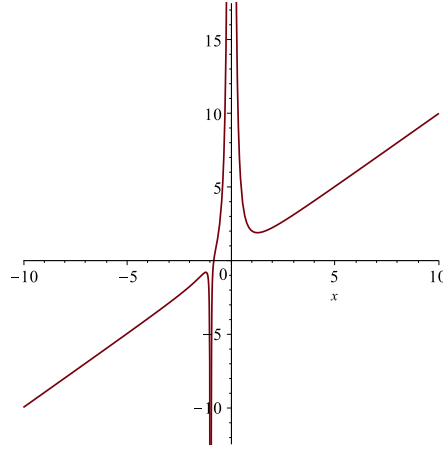


Figure 2.5: Plot of the map $f(x) = x + \frac{\mu}{(x-1+\mu)^2} + \frac{\mu-1}{(x+\mu)^2}$, with $\mu = (1.0123)^{-1}$.

$|Af(\bar{x})|$. To determine Z_2 we make use of r_* . With the standard change of variables, $b := c - \bar{x} \in \overline{B_r(0)}$ for $c \in \overline{B_r(\bar{x})}$ where $0 \leq r \leq r_*$, we use the mean value inequality (Corollary 2.2.6) to obtain

$$\begin{aligned} \max_{c \in \overline{B_r(\bar{x})}} |A(f'(c) - f'(\bar{x}))| &= \max_{b \in \overline{B_r(0)}} A|f'(\bar{x} + b) - f'(\bar{x})| \\ &\leq \max_{b \in \overline{B_r(0)}} \sup_{z \in \overline{B_r(\bar{x})}} A|f''(z)||b| \\ &\leq \left(A \sup_{z \in \overline{B_{r_*}(\bar{x})}} |f''(z)| \right) r =: Z_2 r, \end{aligned}$$

which provides the same bound as the one provided by (2.43). Using the formula

$$f''(x) = \frac{6\mu}{(x-1+\mu)^4} + \frac{6(\mu-1)}{(x+\mu)^4},$$

we can use interval arithmetic to show that we can choose $Z_2 = 15.4$. The resulting radii polynomial is

$$p(r) \stackrel{\text{def}}{=} 15.4r^2 - r + 4 \times 10^{-17}.$$

A direct calculation shows that if $r \in [4.1 \times 10^{-17}, 0.064] \cap [0, r_*] = [4.1 \times 10^{-17}, 0.01]$, then $p(r) < 0$. We conclude the existence of a unique $\tilde{x} \in [\bar{x} - 4 \times 10^{-17}, \bar{x} + 4.1 \times 10^{-17}]$ such that $f(\tilde{x}) = 0$.

The justification for Theorem 2.5.2 was given in terms of functions that are defined on bounded open sets. However, as the following example indicates it can be of use even when the function is defined on all of $\mathbb{R}^n$.

Example 2.5.5. Consider the following model

$$\begin{cases} \dot{x}_1 = 3x_1(1 - x_1) - x_1x_2 - \lambda(1 - e^{-5x_1}) \\ \dot{x}_2 = -x_2 + 3x_1x_2. \end{cases} \quad (2.44)$$

At some given parameter values $\lambda \in \mathbb{R}$, there are up to four real equilibrium solutions.

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the right hand-side of (2.44). Hence,

$$Df(\bar{x}) = \begin{pmatrix} 3 - 6\bar{x}_1 - \bar{x}_2 - 5\lambda e^{-5\bar{x}_1} & -\bar{x}_1 \\ 3\bar{x}_2 & -1 + 3\bar{x}_1 \end{pmatrix}.$$

Since we are working with a two dimensional example, it is easy to obtain an explicit expression for $Df(\bar{x})^{-1}$ and thus we can set $A \stackrel{\text{def}}{=} Df(\bar{x})^{-1}$, so that Z_0 satisfying (2.15) can be taken to be 0. To apply the radii polynomial approach we compute the bounds Y_0 and Z_2 . The bound $Y_0 \geq \|Af(\bar{x})\|_\infty$ can be easily be computed. To compute the bound Z_2 , we use formula (2.43), which in this case is given by

$$Z_2 = Z_2(r_*) = \sup_{b \in B_{r_*}(\bar{x})} \left(\max_{1 \leq i \leq 2} \sum_{k,m=1}^2 \left| \sum_{j=1}^2 A_{ij} \frac{\partial^2 f_j}{\partial x_m \partial x_k}(b) \right| \right).$$

Since the only nonzero second partial derivatives are

$$\frac{\partial^2 f_1}{\partial x_1^2} = -6 + 25\lambda e^{-5x_1}, \quad \frac{\partial^2 f_1}{\partial x_1 \partial x_2} = -1 \quad \text{and} \quad \frac{\partial^2 f_2}{\partial x_1 \partial x_2} = 3,$$

one has that

$$\begin{aligned} Z_2(r_*) &= \sup_{b \in B_{r_*}(\bar{x})} \left(\max_{1 \leq i \leq 2} \sum_{k,m=1}^2 \left| A_{i1} \frac{\partial^2 f_1}{\partial x_m \partial x_k}(b) + A_{i2} \frac{\partial^2 f_2}{\partial x_m \partial x_k}(b) \right| \right) \\ &= \max_{1 \leq i \leq 2} \left\{ \sup_{b_1 \in [\bar{x}_1 - r_*, \bar{x}_1 + r_*]} \left(\left| A_{i1} (6 + 25\lambda e^{-5b_1}) \right| + 2 \left| -A_{i1} + 3A_{i2} \right| \right) \right\}, \end{aligned}$$

and therefore, we set

$$p(r) \stackrel{\text{def}}{=} Z_2(r_*)r^2 - r + Y_0, \quad (2.45)$$

which can only be interpreted for $r \leq r_*$. In particular, Theorem 2.4.2 can be applied to (2.45) for $r_0 \leq r_*$ only.

To demonstrate the use of the quadratic radii polynomial (2.45) we fix $\lambda = 1/2$ and consider $\bar{x} = (\bar{x}_1, \bar{x}_2) \in \mathbb{R}^2$ such that $\|f(\bar{x})\| < \text{tol}$ for some fixed tolerance `tol`. In this example, we choose `tol` = 10^{-15} , and compute four approximate solutions with Newton's method to obtain

$$\begin{aligned} \bar{x}^{(1)} &= \begin{pmatrix} 0.7940710596186642 \\ 0 \end{pmatrix}, & \bar{x}^{(2)} &= \begin{pmatrix} 0.3333333333333334 \\ 0.7833134042563428 \end{pmatrix}, \\ \bar{x}^{(3)} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}, & \bar{x}^{(4)} &= \begin{pmatrix} -0.1095121750607109 \\ 0 \end{pmatrix}. \end{aligned} \quad (2.46)$$

For each $i = 1, 2, 3, 4$, we fixed $r_* = 0.2$, defined the corresponding radii polynomial $p^{(i)}$ and computed an interval $I^{(i)}$ such that $I^{(i)} \subset \{r \mid p^{(i)}(r) < 0\} \cap [0, r_*] = \text{El}(p^{(i)})$, that is

$$\begin{aligned} I^{(1)} &= [1.6 \times 10^{-16}, 0.2] \\ I^{(2)} &= [1.1 \times 10^{-15}, 0.0265] \\ I^{(3)} &= (0, 0.017] \\ I^{(4)} &= [6 \times 10^{-16}, 0.015]. \end{aligned} \tag{2.47}$$

Notice that the value $r_* = 0.2$ restricts the existence interval for $\bar{x}^{(1)}$ but not for the other equilibria. It is in fact an interesting problem to optimize the size of the existence interval by playing with different values of r_* (see Exercise 2.7.5).

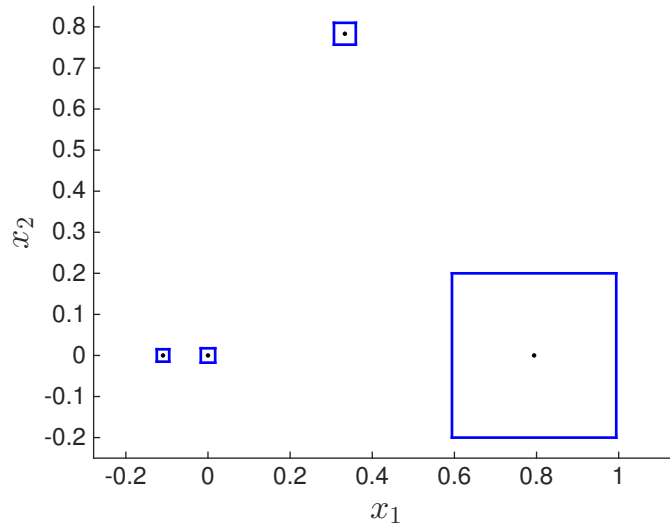


Figure 2.6: Largest existence and uniqueness enclosures for each equilibria of (2.44) for $\lambda = 1/2$. For each $i = 1, 2, 3, 4$, the radius around $\bar{x}^{(i)}$ is the largest value of $I^{(i)}$. The quantities $\bar{x}^{(i)}$ and $I^{(i)}$ are found in (2.46) and (2.47) respectively.

Let us make some observation about the examples discussed above. In the case of a quadratic nonlinearity (Examples 2.4.5, 2.4.6, 2.4.8, 2.4.9 and 2.4.15) the bound Z_2 reduced to a constant (independent of r) automatically. In the case of a cubic nonlinearity (Examples 2.4.7) the bound Z_2 is linear in r and the resulting radii polynomial is cubic. As the last examples demonstrate, for non polynomial nonlinearities we may sometimes obtain a transcendental function $p(r)$, which is then “reduced” to a quadratic polynomial by putting an a priori restriction r_* on r .

2.6 Effectiveness of Radii Polynomials

Theorem 2.5.2 provides, in the form of a radii polynomial, a sufficient condition to prove the existence of a zero of a function. The following theorem demonstrates that if a zero is nondegenerate, then it is always possible to construct a radii polynomial that will guarantee the existence of the zero.

Theorem 2.6.1. *Let $U \subset \mathbb{R}^n$ be an open set and let $f \in C^2(U, \mathbb{R}^n)$. If $\tilde{x} \in U$ is a nondegenerate zero of f , then there exists $\epsilon > 0$ such that if $\bar{x} \in \overline{B_\epsilon(\tilde{x})}$ then the following holds. There exists $A(\bar{x}) \in M_n(\mathbb{R})$, $r_*(\bar{x})$, $Y_0(\bar{x})$, $Z_0(\bar{x})$, and $Z_2(\bar{x})$ satisfying (2.38), (2.39), and (2.40), respectively, and positive constants $r_0(\bar{x}) < r_*(\bar{x})$ such that*

$$p(r_0(\bar{x})) < 0$$

where

$$p(r) \stackrel{\text{def}}{=} Z_2(\bar{x})r^2 - (1 - Z_0(\bar{x}))r + Y_0(\bar{x}) \quad (2.48)$$

is the associated radii polynomial and $\overline{B_{r_*(\bar{x})}(\bar{x})} \subset U$.

Proof. The first step in the proof is to identify an approximate zero $\bar{x}$. Choose $\delta_0 > 0$ such that $\overline{B_{\delta_0}(\tilde{x})} \subset U$. Choose $0 < \delta_1 < \delta_0$ such that $Df(x)$ is invertible for $x \in \overline{B_{\delta_1}(\tilde{x})}$ and

$$\sup_{x \in \overline{B_{\delta_1}(\tilde{x})}} \|Df(x)^{-1}\| \leq 2\|Df(\tilde{x})^{-1}\| \quad (2.49)$$

and set $r_* = \delta_1$. Let

$$M := \sup_{x \in \overline{B_{\delta_1}(\tilde{x})}} \|D^2f(x)\|.$$

Choose $0 < \delta_2 < \delta_1/4$ such that

$$\sup_{x \in \overline{B_{\delta_2}(\tilde{x})}} \|f(x)\| < \frac{1}{128\|Df(\tilde{x})^{-1}\|^2 M}.$$

Choose $0 < \delta_3 < \delta_2$ such that

$$\sup_{x \in \overline{B_{\delta_3}(\tilde{x})}} \|Df(x) - Df(\tilde{x})\| < \frac{1}{8\sqrt{2}\|Df(\tilde{x})^{-1}\|}.$$

Choose $\hat{x} \in \overline{B_{\delta_3}(\tilde{x})}$ and $A \in M_n(\mathbb{R})$ such that

$$\|I - ADf(\hat{x})\| \leq \frac{1}{4} \quad \text{and} \quad \|A\| \leq \sqrt{2}\|Df(\hat{x})^{-1}\|.$$

Finally, let $0 < \delta_4 < \delta_3$ and consider $\bar{x} \in \overline{B_{\delta_4}(\tilde{x})}$. Note that we have not yet fixed the value of δ_4 as it will need to be chosen sufficiently small.

We now show that we can use Theorem 2.5.2 to prove the existence of $\tilde{x}$. Observe that

$$\begin{aligned}
\|Id - Adf(\tilde{x})\| &\leq \|Id - Adf(\dot{x})\| + \|Adf(\dot{x}) - Adf(\tilde{x})\| \\
&\leq \frac{1}{4} + \|A\| \|Df(\dot{x}) - Df(\tilde{x})\| \\
&\leq \frac{1}{4} + \sqrt{2} \|Df(\dot{x})^{-1}\| \frac{1}{8\sqrt{2} \|Df(\tilde{x})^{-1}\|} \\
&\leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2},
\end{aligned}$$

where the last inequality follows from (2.49) and the fact that $\dot{x} \in \overline{B_{\delta_1}(\tilde{x})}$.

Set

$$\begin{aligned}
Y_0 &\stackrel{\text{def}}{=} \|A\| \|f(\tilde{x})\| \\
Z_0 &\stackrel{\text{def}}{=} \frac{1}{2} \\
Z_2 &\stackrel{\text{def}}{=} \|A\| M.
\end{aligned}$$

We make two observations. First,

$$\lim_{\delta_4 \rightarrow 0} Y_0 = 0, \tag{2.50}$$

and second,

$$\begin{aligned}
Y_0 &= \|A\| \|f(\tilde{x})\| \\
&< \|A\| \frac{1}{128 \|Df(\tilde{x})^{-1}\|^2 M} \\
&= \left(\frac{\|A\|}{\|Df(\tilde{x})^{-1}\|} \right)^2 \frac{1}{128 \|A\| M} \\
&\leq \left(\frac{\sqrt{2} \|Df(\dot{x})^{-1}\|}{\|Df(\tilde{x})^{-1}\|} \right)^2 \frac{1}{128 Z_2} \\
&\leq \frac{1}{16 Z_2} \\
Y_0 Z_2 &< \frac{1}{16}
\end{aligned}$$

where the second to last inequality follows from (2.49) and the fact that $\dot{x} \in \overline{B_{\delta_1}(\tilde{x})}$.

By definition the radii polynomial is

$$p(r) = Z_2 r^2 - (1 - Z_0) r + Y_0 = Z_2 r^2 - \frac{1}{2} r + Y_0$$

and hence the discriminant of the radii polynomial is

$$\Delta = \frac{1}{4} - 4Y_0 Z_2 > 0.$$

Set

$$r_- = \frac{1}{2Z_2} \left(1/2 - \sqrt{\Delta} \right) \quad \text{and} \quad r_+ = \min \left\{ \frac{1}{2Z_2} \left(1/2 + \sqrt{\Delta} \right), r_* \right\}.$$

By (2.50) if δ_4 is sufficiently small, then $r_- < r_*$. With such a choice of δ_4 , if $r_0 \in (r_-, r_+)$, then $p(r_0) < 0$, and the result follows. $\square$

2.7 Exercises

Exercise 2.7.1. Prove that if $A: \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a linear map and $\|I - A\| < 1$, then A is invertible.

Exercise 2.7.2. Let $x \in \mathbb{R}^N$ and let $\nu > 0$. Define

$$\|x\|_{1,\nu} := \sum_{n=1}^N |x_n| \nu^n.$$

Prove that $\|\cdot\|_{1,\nu}$ is a norm and that the associate matrix norm is given by

$$\|A\|_{\nu,N} = \max_{1 \leq j \leq N} \frac{1}{\nu^j} \sum_{0 \leq i \leq N} |A_{i,j}| \nu^i,$$

for $A = [a_{i,j}] \in M_N(\mathbb{C})$.

Exercise 2.7.3. Prove Proposition 2.4.11.

Exercise 2.7.4. Prove Remark 2.5.3, that is show that the bound Z_2 as defined in (2.43) satisfies (2.40).

Exercise 2.7.5. In Example 2.5.5, we computed rigorously four distinct solutions of $f(x, y) = (0, 0)$, where

$$f(x, y) = \begin{pmatrix} 3x(1-x) - xy - \frac{1}{2}(1 - e^{-5x}) \\ -y + 3xy \end{pmatrix}.$$

We fixed $r_* = 0.1$ and found that the existence interval were given by (2.47). For each $\bar{x}^{(i)}$ ($i = 1, 2, 3, 4$), vary the value of r_* in order to find the largest possible r_+ for which the associated radii polynomial is negative.

Exercise 2.7.6. Assume $f \in C^2(\mathbb{R}^n, \mathbb{R}^n)$ and that $\tilde{x}$ is a nondegenerate zero of f . Consider the Newton operator $T(x) = x - Df(x)^{-1}f(x)$. Prove that there exists $r > 0$ such that if $x^0 \in B_r(\tilde{x})$ and $x^{n+1} = T^n(x^0)$, then the sequence x^n converges to $\tilde{x}$ at a *quadratic rate*, that is, if

$$\epsilon_n = \|f(x_n)\|,$$

then

$$\epsilon_{n+1} \leq M \epsilon_n^2,$$

for some $M > 0$.

Exercise 2.7.7. Prove the following important corollary of the mean value theorem.

Corollary 2.7.8. *Suppose $U \subset \mathbb{R}^n$ is open and that $f: U \rightarrow \mathbb{R}^m$ is C^1 . Then f is locally Lipschitz.*

Exercise 2.7.9. Use the radii polynomial approach to give rigorous bounds on the value of $\sqrt{7}$.

Exercise 2.7.10. Use the radii polynomial approach to give rigorous bounds on the zeros of $f(x) = xe^x - 1$.

Exercise 2.7.11. Use Newton's method in the complex variable setting to obtain numerical approximations of the three zeros of $z^3 + z + 1 = 0$. Use the radii polynomial approach to provide rigorous bounds on the values of the true zeros.

Exercise 2.7.12. Consider the FitzHugh-Nagumo equation

$$\dot{x} = f(x) = \begin{pmatrix} x_1(x_1 - a)(1 - x_1) - x_2 \\ \epsilon(x_1 - \gamma x_2) \end{pmatrix} \quad (2.51)$$

at the parameter values $(a, \epsilon, \gamma) = (5, 1, 2)$. Consider the initial point $x^0 = (1, 0)^T \in \mathbb{R}^2$ and apply Newton's method to converge to an approximate steady states $\bar{x} \in \mathbb{R}^2$. What is the point $\bar{x}$ you obtain? Decrease the parameter γ by .05 and apply Newton's method with initial point $x^0 = \bar{x}$, where $\bar{x}$ is the value obtained at the previous step. This initial guess is called a 0th order predictor. Applying Newton's method, you should converge to a new point that we denote again by $\bar{x}$. Repeat this procedure 40 times. What do you observe? This procedure is the simplest form of a predictor-corrector method. Then, for each of the numerical approximations, use the radii polynomial approach to prove your computations.

Exercise 2.7.13. As presented in more details in Section 13.4, the following example arises when studying the steady states of the diblock-copolymer partial differential equation model. Consider a number $N \in \mathbb{N}$ (the finite dimensional projection), and define $f: \mathbb{R}^N \rightarrow \mathbb{R}^N$ component-wise by

$$f_k(x) = (-k^4 + \lambda k^2 - \lambda \sigma) x_k - \lambda k^2 \sum_{\substack{k_1 + k_2 + k_3 = k \\ k_i = -N, \dots, N}} x_{|k_1|} x_{|k_2|} x_{|k_3|}, \quad \text{for } k = 1, \dots, N, \quad (2.52)$$

where in the discrete convolution, it is assumed that $x_0 = 0$, and where $\lambda, \sigma \in \mathbb{R}$ are parameters.

For example the case $N = 2$ leads to the two-dimensional map

$$f(x) = \begin{pmatrix} (-1 + \lambda - \lambda \sigma) x_1 - \lambda (3x_1^3 + 6x_1 x_2^2) \\ (-16 + 4\lambda - \lambda \sigma) x_2 - 4\lambda (6x_1^2 x_2 + 3x_2^3) \end{pmatrix}, \quad (2.53)$$

while the case $N = 3$ leads to the three-dimensional map

$$f(x) = \begin{pmatrix} (-1 + \lambda - \lambda\sigma)x_1 - \lambda(3x_1^3 + 3x_1^2x_3 + 6x_1x_2^2 + 6x_1x_2^2x_3 + 3x_2^2x_3) \\ (-16 + 4\lambda - \lambda\sigma)x_2 - 4\lambda(6x_1^2x_2 + 6x_1x_2x_3 + 3x_2^3 + 6x_3x_2^2) \\ (-81 + 9\lambda - \lambda\sigma)x_3 - 9\lambda(x_1^3 + 6x_1^2x_3 + 3x_1x_2^2 + 6x_2^2x_3 + 3x_3^3) \end{pmatrix}. \quad (2.54)$$

Our goal is to determine a radii polynomial for arbitrary $N \in \mathbb{N}$. We assume that we are given $\bar{x} \in \mathbb{R}^N$ such that $f(\bar{x}) \approx 0$ and that we can compute $A \in M_N(\mathbb{R})$ such that $A \approx Df(\bar{x})^{-1}$.

We use the weighted 1-norm in $\mathbb{R}^N$ given by $\|x\|_1 = 2 \sum_{k=1}^N |x_k|$ and leave it as an exercise to check that

$$\|x *_e y\|_1 \leq \|x\|_1 \|y\|_1. \quad (2.55)$$

We can compute the following bounds

$$\begin{aligned} Y_0 &= \|Af(\bar{x})\|_1 \\ Z_0 &= \|I - ADf(\bar{x})\|_1. \end{aligned}$$

To obtain Z_2 , for $c \in \overline{B_r(\bar{x})}$ and $h \in \overline{B_1(0)} \subset \mathbb{R}^N$, we investigate

$$\begin{aligned} ([Df(c) - Df(\bar{x})]h)_k &= (-k^4 + \lambda k^2 - \lambda\sigma)h_k - 3\lambda k^2(c * c * h)_k \\ &\quad - (-k^4 + \lambda k^2 - \lambda\sigma)h_k + 3\lambda k^2(\bar{x} * \bar{x} * h)_k \\ &= -3\lambda k^2((c * c - \bar{x} * \bar{x}) * h)_k \\ &= -3\lambda k^2((c + \bar{x}) * (c - \bar{x}) * h)_k. \end{aligned}$$

Denote by $\Lambda \in M_N(\mathbb{R})$ the diagonal matrix defined by $(\Lambda x)_k = k^2 x_k$. Hence, using (2.55),

$$\begin{aligned} \|A[Df(c) - Df(\bar{x})]\|_1 &= \sup_{\|h\|_1=1} \|A[Df(c) - Df(\bar{x})]h\|_1 \\ &\leq 3|\lambda| \|A\Lambda((c + \bar{x}) * (c - \bar{x}) * h)\|_1 \\ &\leq 3|\lambda| \|A\Lambda\|_1 \|c + \bar{x}\|_1 \|c - \bar{x}\|_1 \|h\|_1 \\ &\leq 3|\lambda| \|A\Lambda\|_1 (\|c\|_1 + \|\bar{x}\|_1)r \\ &\leq 3|\lambda| \|A\Lambda\|_1 (2\|\bar{x}\|_1 + r)r, \end{aligned}$$

and therefore, we set

$$Z_2 = 3|\lambda| \|A\Lambda\|_1 (2\|\bar{x}\|_1 + r)$$

so that the radii polynomial is given by

$$p(r) = (3|\lambda| \|A\Lambda\|_1) r^3 + (6|\lambda| \|\bar{x}\|_1 \|A\Lambda\|_1) r^2 - (1 - Z_0)r + Y_0.$$

We fixed the parameters $N = 10$, $\lambda = 10$ and $\sigma = 2$, and applied Newton's method to find $\bar{x} \in \mathbb{R}^{10}$ such that $\bar{x}_2 = 0.183635760099768$, $\bar{x}_6 = -0.002167639989372$ and $\bar{x}_{10} = 0.000023497621288$ with all other components equal to 0. We found that

$$I = [1 \times 10^{-14}, 0.077] \subset \text{El}(p).$$

which implies that there exists $\tilde{x} \in \mathbb{R}^{10}$ with $\|\tilde{x} - \bar{x}\|_1 = 2 \sum_{k=1}^{10} |\tilde{x}_k - \bar{x}_k| \leq 1 \times 10^{-14}$ such that $f(\tilde{x}) = 0$.

Exercise 2.7.14. In this exercise, we consider the particular case of the restricted four body problem where three point masses move in circular periodic orbits around their center of mass and a fourth massless particle interacts with this system without affecting their motion. This particular case is known as the *Circular Equilateral Restricted Four-Body Problem* (CR4BP). The equations of motion of the massless particle in dimensionless coordinates relative to a reference (synodic) frame can be written as

$$\begin{aligned}\ddot{x} - 2\dot{y} &= \Omega_x, \\ \ddot{y} + 2\dot{x} &= \Omega_y, \\ \ddot{z} &= \Omega_z,\end{aligned}$$

where

$$\Omega(x, y, z, m_1, m_2, m_3) \stackrel{\text{def}}{=} \frac{1}{2}(x^2 + y^2) + \sum_{i=1}^3 \frac{m_i}{r_i}, \quad (2.56)$$

is called the *effective potential*, and $r_i \stackrel{\text{def}}{=} \sqrt{(x - x_i)^2 + (y - y_i)^2 + z^2}$, for $i = 1, 2, 3$. The general expressions of the coordinates of the three *primaries* in terms of the masses of the three point masses are given by

$$\begin{aligned}(x_1, y_1, z_1) &= \left(\frac{-|K|\sqrt{m_2^2 + m_2m_3 + m_3^2}}{K}, 0, 0 \right) \\ (x_2, y_2, z_2) &= \left(\frac{|K|[(m_2 - m_3)m_3 + m_1(2m_2 + m_3)]}{2K\sqrt{m_2^2 + m_2m_3 + m_3^2}}, \frac{-\sqrt{3}m_3}{2m_2^{3/2}}\sqrt{\frac{m_2^3}{m_2^2 + m_2m_3 + m_3^2}}, 0 \right) \\ (x_3, y_3, z_3) &= \left(\frac{|K|}{2\sqrt{m_2^2 + m_2m_3 + m_3^2}}, \frac{\sqrt{3}}{2\sqrt{m_2}}\sqrt{\frac{m_2^3}{m_2^2 + m_2m_3 + m_3^2}}, 0 \right)\end{aligned}$$

where $K \stackrel{\text{def}}{=} m_2(m_3 - m_2) + m_1(m_2 + 2m_3)$ and the three masses satisfy the relation $m_1 + m_2 + m_3 = 1$.

The equilibria of the system are given by the critical points of the effective potential $\Omega(x, y, z, m_1, m_2, m_3)$ given by (2.56), that is they satisfy the equations $\Omega_x = 0$, $\Omega_y = 0$ and $\Omega_z = 0$. A straightforward computation shows that

$$\Omega_x = x - \sum_{i=1}^3 \frac{m_i}{r_i^3}(x - x_i), \quad \Omega_y = y - \sum_{i=1}^3 \frac{m_i}{r_i^3}(y - y_i), \quad \Omega_z = -z \sum_{i=1}^3 \frac{m_i}{r_i^3},$$

and therefore as a consequence all the equilibrium points are coplanar, since $\Omega_z = 0 \Rightarrow z = 0$. Hence, the relative equilibria $(x, y, 0)$ satisfy the equation

$$F(x, y) \stackrel{\text{def}}{=} \begin{pmatrix} x - \sum_{i=1}^3 \frac{m_i(x - x_i)}{\left(\sqrt{(x - x_i)^2 + (y - y_i)^2}\right)^3} \\ y - \sum_{i=1}^3 \frac{m_i(y - y_i)}{\left(\sqrt{(x - x_i)^2 + (y - y_i)^2}\right)^3} \end{pmatrix} = 0 \quad (2.57)$$

Using the (localized) radii polynomial approach as proposed in Theorem 2.5.2, prove that Equation (2.57) has at least ten equilibrium points for the case $m_1 = m_2 = m_3 = 1/3$ and eight equilibrium points for the non equal masses case $m_1 = 0.9987451087$, $m_2 = 0.0010170039$ and $m_3 = 0.0002378873$. To obtain your results, use the bound Z_2 as defined in (2.43).

Discrete Time Dynamics

Exercise 2.7.15.

Consider the Logistic map $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \mu x(1 - x)$, where $\mu \in \mathbb{R}$ is a parameter. Compute and prove (using the radii polynomial approach) the following:

- A period-2 point at the parameter value $\mu = 3.2$.
- A period-3 point at the parameter value $\mu = 3.83$.
- A period-3 point at the parameter value $\mu = 3.85$.

If you succeeded in this exercise, you have proven the existence of chaos in the Logistic map at the parameter values $\mu = 3.83$ and $\mu = 3.85$. Indeed, a famous result in discrete dynamical systems states that if there exists a period-3 point for a continuous interval map $F: I \rightarrow I$ (where $I \subset \mathbb{R}$ is a closed interval), then the map F has periodic points of *all* periods and even exhibits chaotic dynamics [22, 36]. This is a nice example of what is sometimes called a *forcing theorem*: proving the existence of some solutions *forces* the existence of other solutions.

Exercise 2.7.16. Consider the Hénon map $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$F(x) = \begin{pmatrix} 1 - ax_1^2 + x_2 \\ bx_1 \end{pmatrix},$$

where $a, b \in \mathbb{R}$ are parameters. Compute and prove (using the radii polynomial approach) the following:

- A period-2 point at the parameter values $a = b = \frac{1}{2}$.
- A period-4 point at the parameter value $a = 1$ and $b = 0.3$.

Exercise 2.7.17.

Consider the equation $z^3 - 1 = 0$. Write Newton's method for solving this equation as

$$z_{n+1} = F(z_n),$$

where $F: \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C}$ is given by

$$F(z) = \frac{2z^3 + 1}{3z^2}.$$

Compute and prove (using the radii polynomial approach) the following

- A period - 3 point.
- A period - 5 point.
- Try to find points of higher period. Say points with all periods up to 100. Plot all these periodic orbits in the plane on the same graph. Prove you have all of these orbits using the radii polynomial approach.

